

# BFS Traversal

Step 1: Start at A (Seeking target node K)

## Legend



Complete



Processing



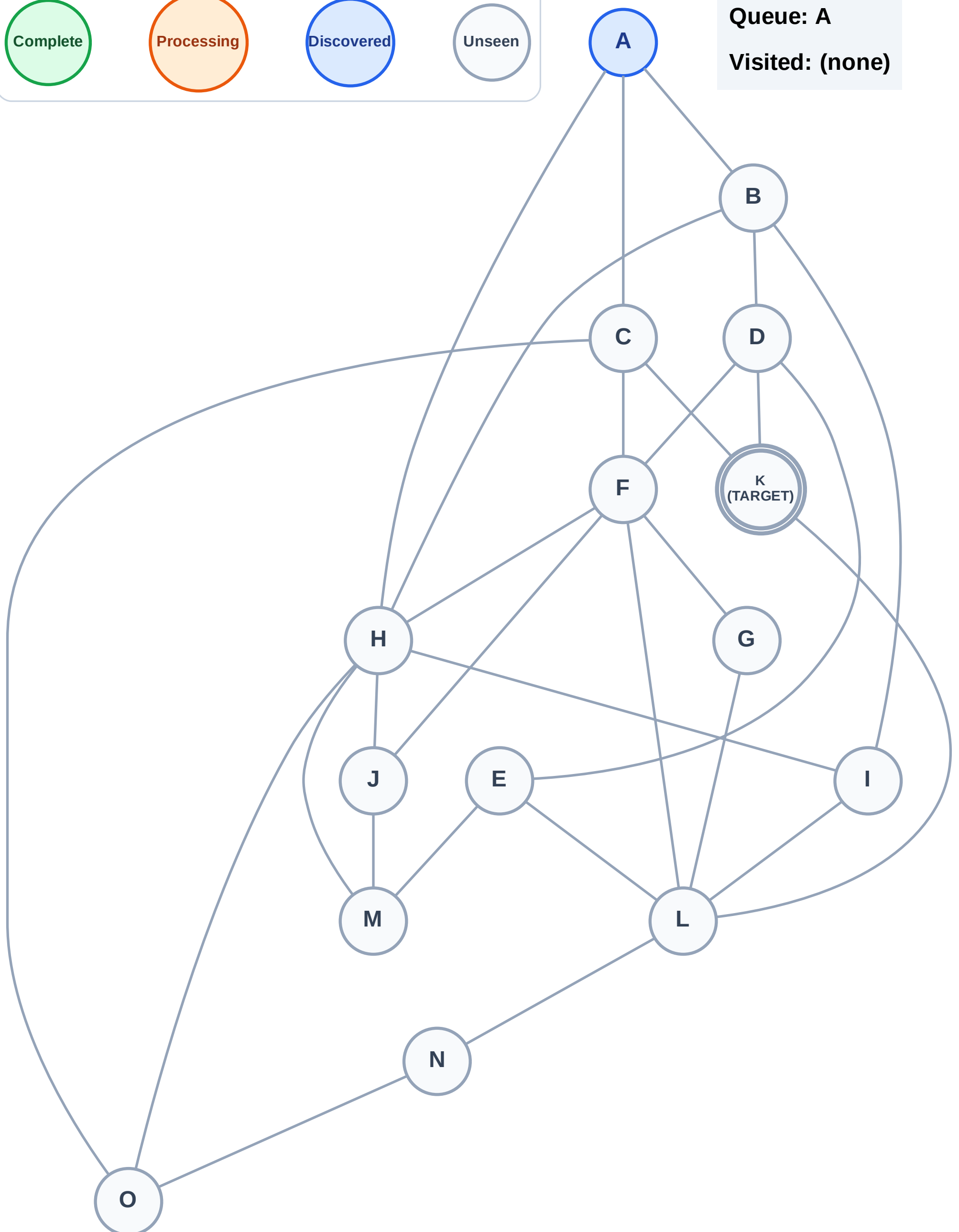
Discovered



Unseen

Queue: A

Visited: (none)



# BFS Traversal

Step 2: Process node A

Legend

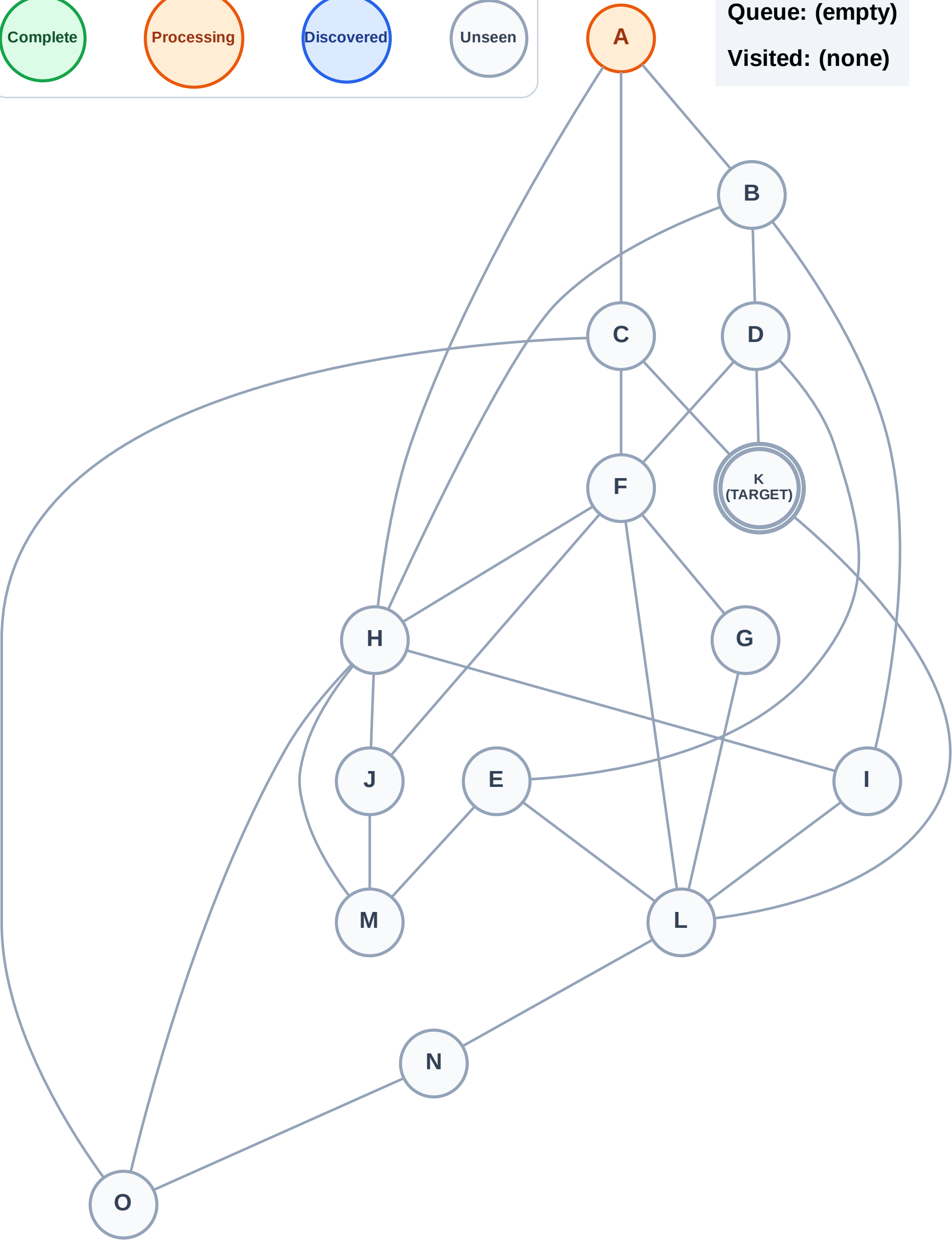
Complete

Processing

Discovered

Unseen

Queue: (empty)  
Visited: (none)



# BFS Traversal

Step 3: Discover B, C, H from A

## Legend



Complete



Processing



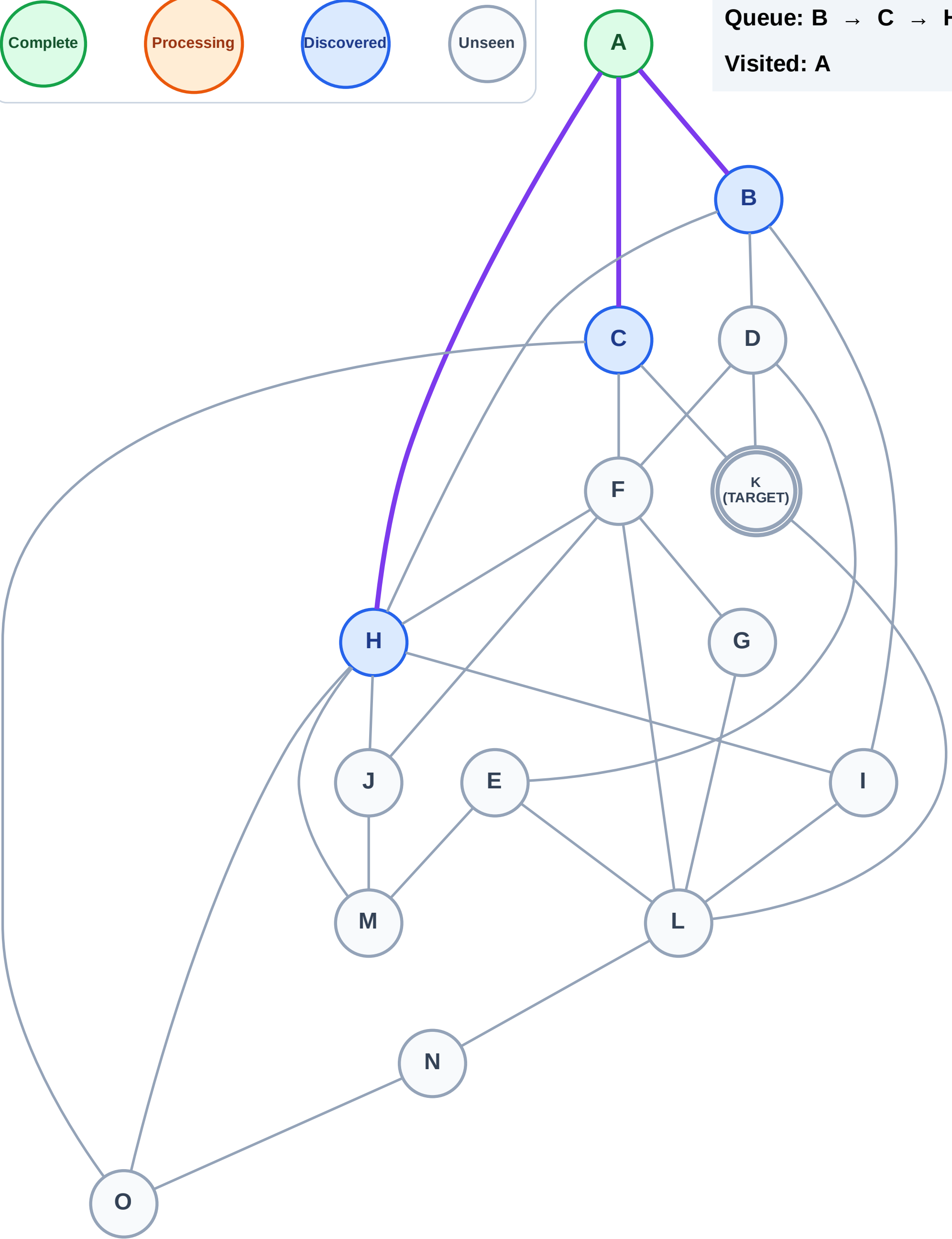
Discovered



Unseen

Queue: B → C → H

Visited: A



# BFS Traversal

Step 4: Process node B

Legend

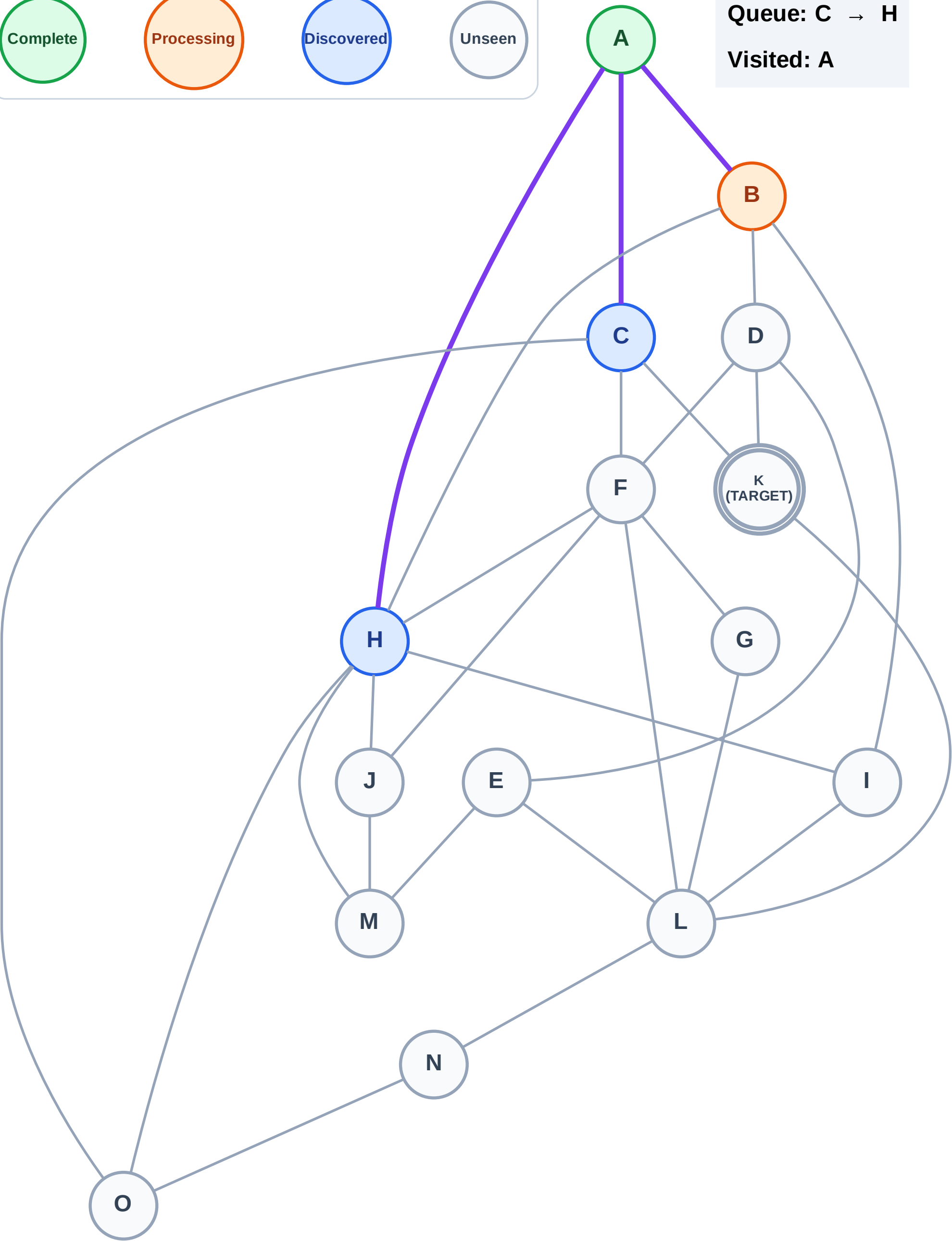
Complete

Processing

Discovered

Unseen

Queue: C → H  
Visited: A



# BFS Traversal

Step 5: Discover D, I from B

Legend

Complete

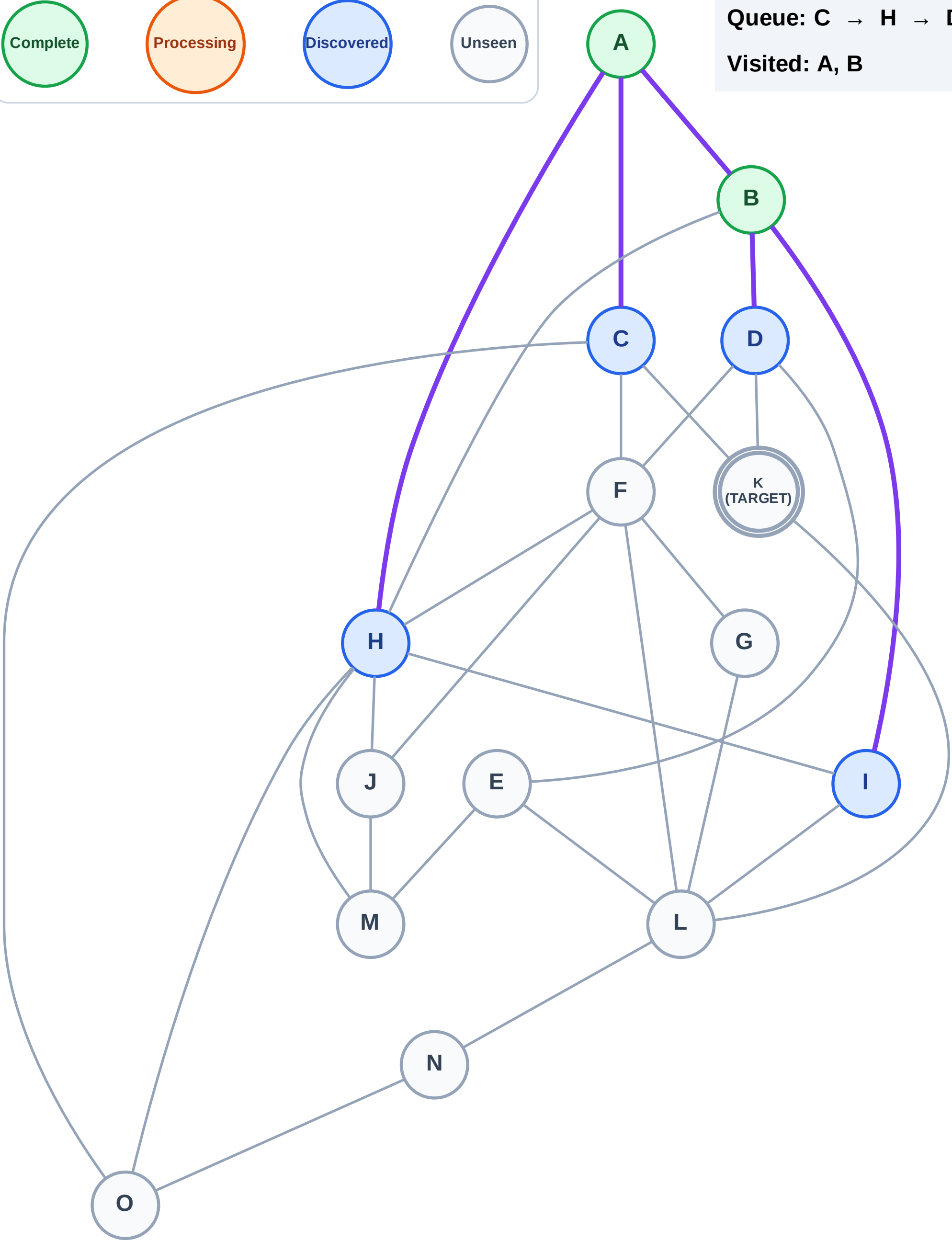
Processing

Discovered

Unseen

Queue: C → H → D → I

Visited: A, B



# BFS Traversal

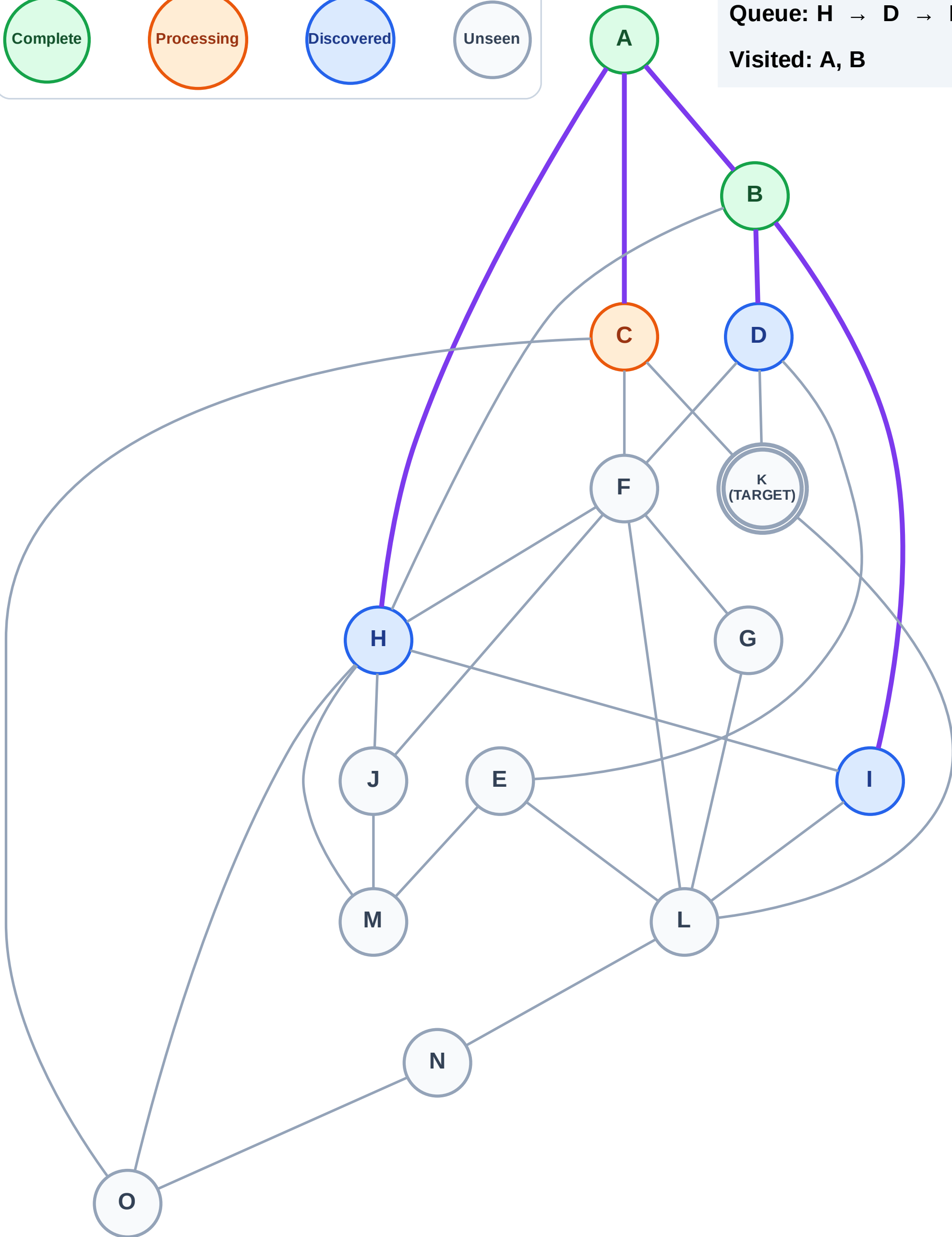
Step 6: Process node C

## Legend



Queue: H → D → I

Visited: A, B

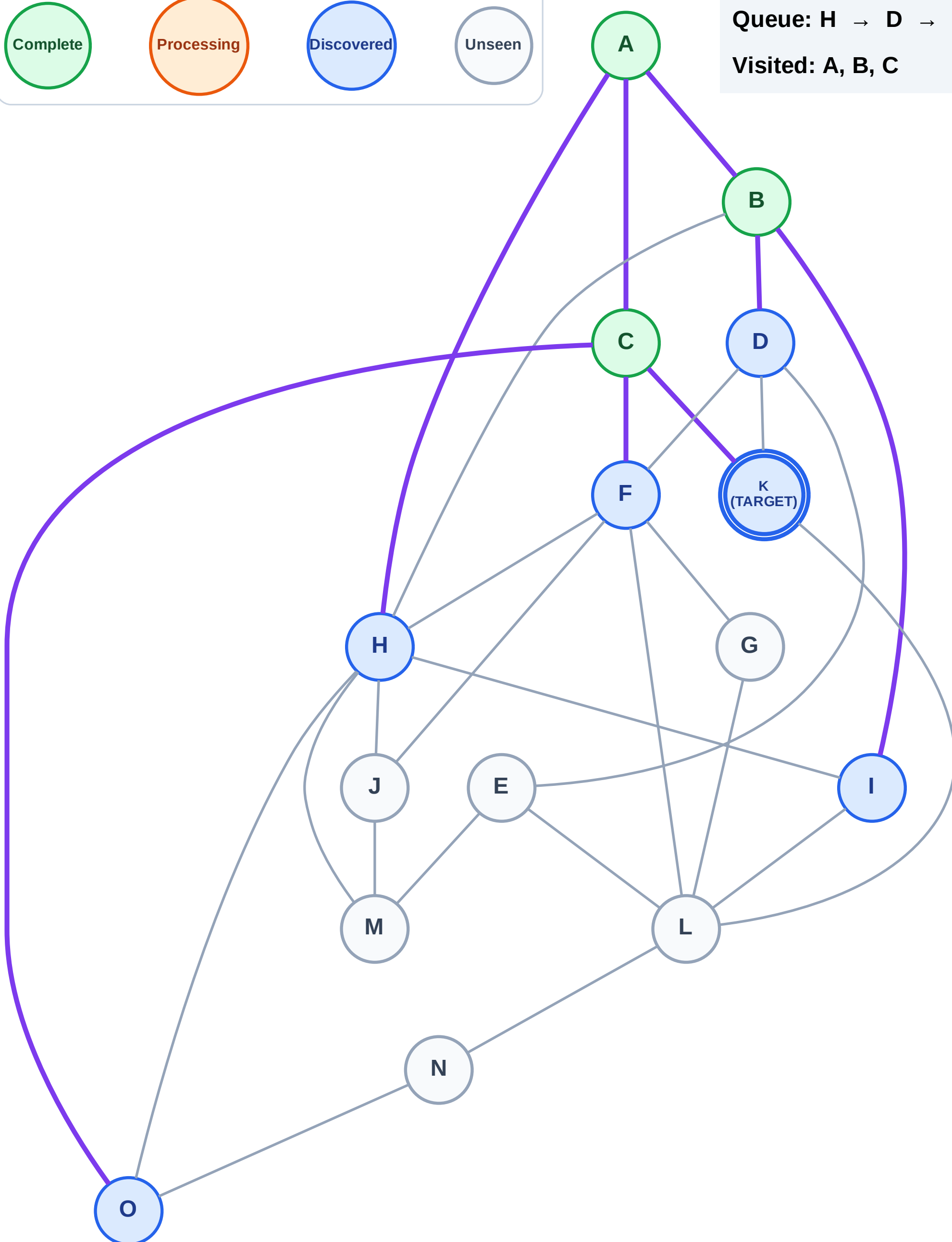


### Step 7: Discover F, K, O from C

### Legend

## Unseen

**Visited: A, B, C**



# BFS Traversal

Step 8: Process node H

Legend

Complete

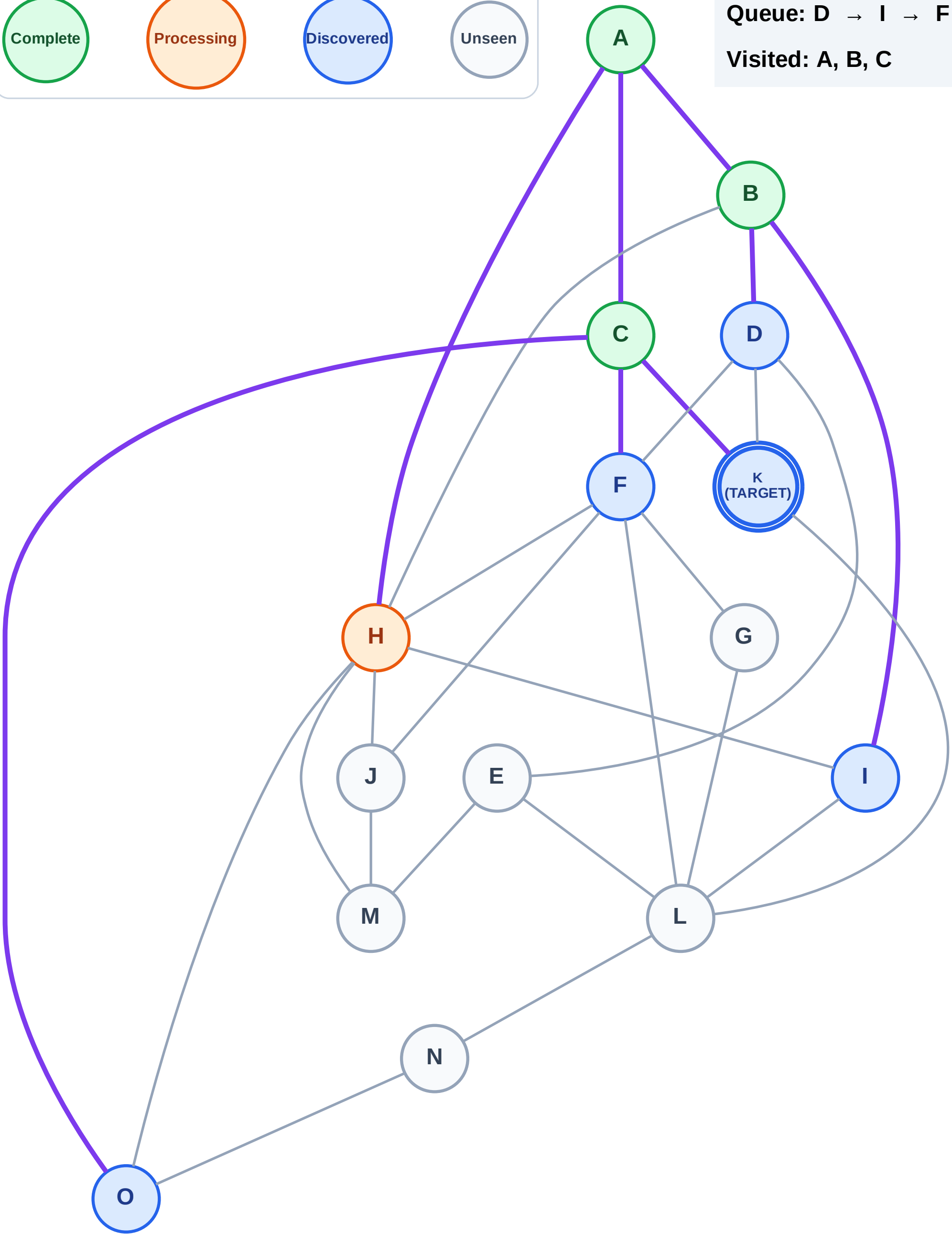
Processing

Discovered

Unseen

Queue: D → I → F → K → O

Visited: A, B, C



# BFS Traversal

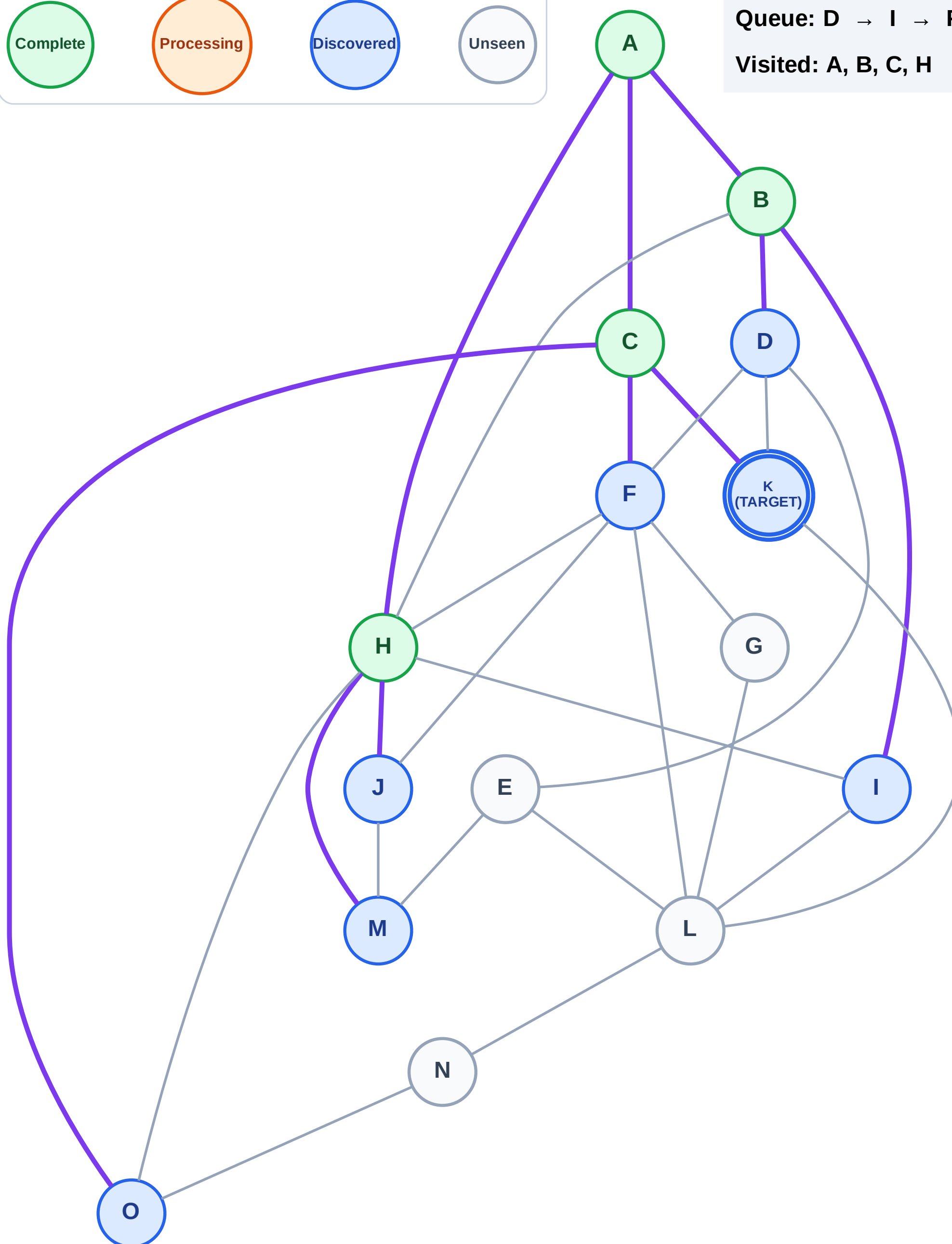
Step 9: Discover J, M from H

## Legend



Queue: D → I → F → K → O → J → M

Visited: A, B, C, H



# BFS Traversal

Step 10: Process node D

Legend

Complete

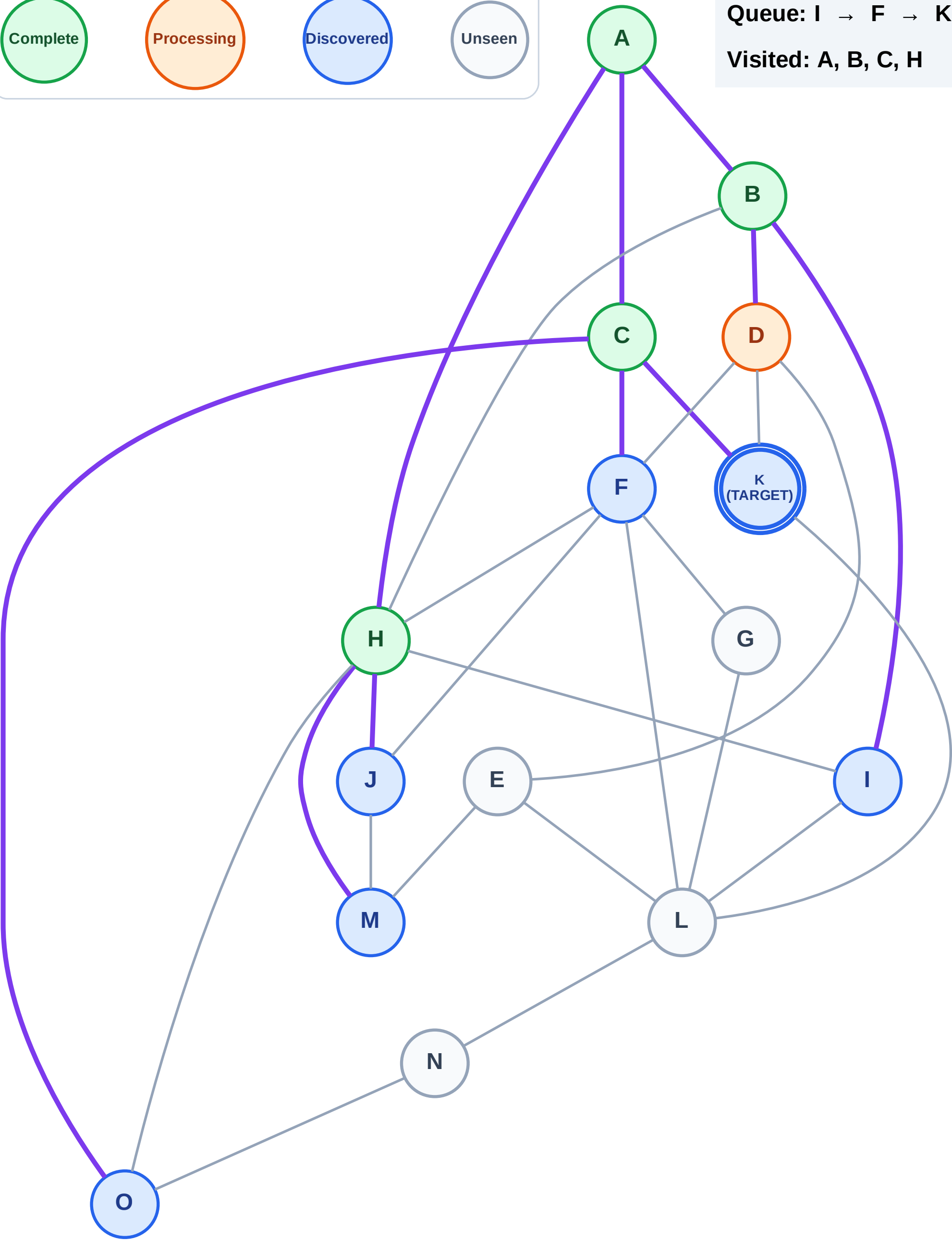
Processing

Discovered

Unseen

Queue: I → F → K → O → J → M

Visited: A, B, C, H



# BFS Traversal

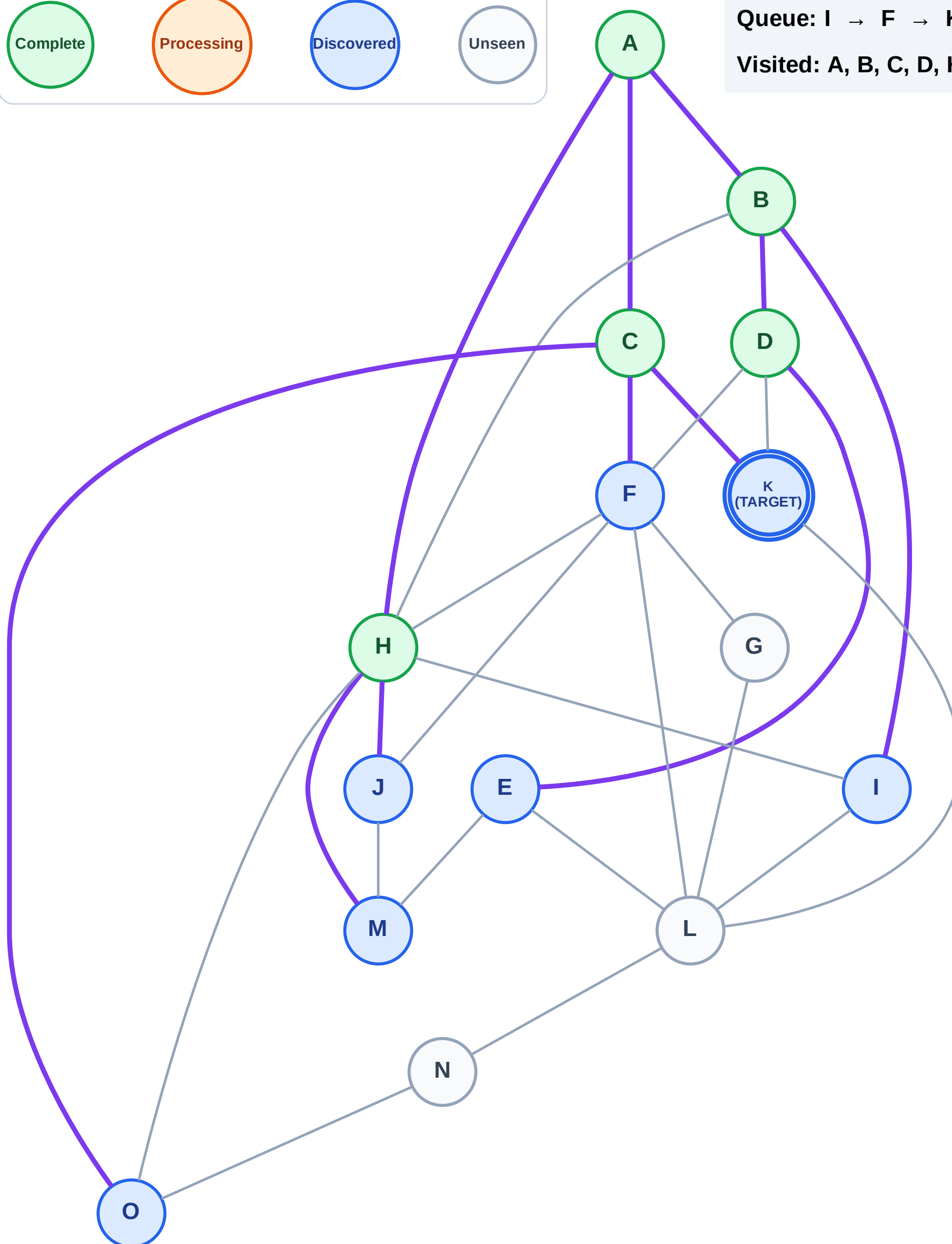
Step 11: Discover E from D

## Legend



Queue: I → F → K → O → J → M → E

Visited: A, B, C, D, H



# BFS Traversal

Step 12: Process node I

Legend

Complete

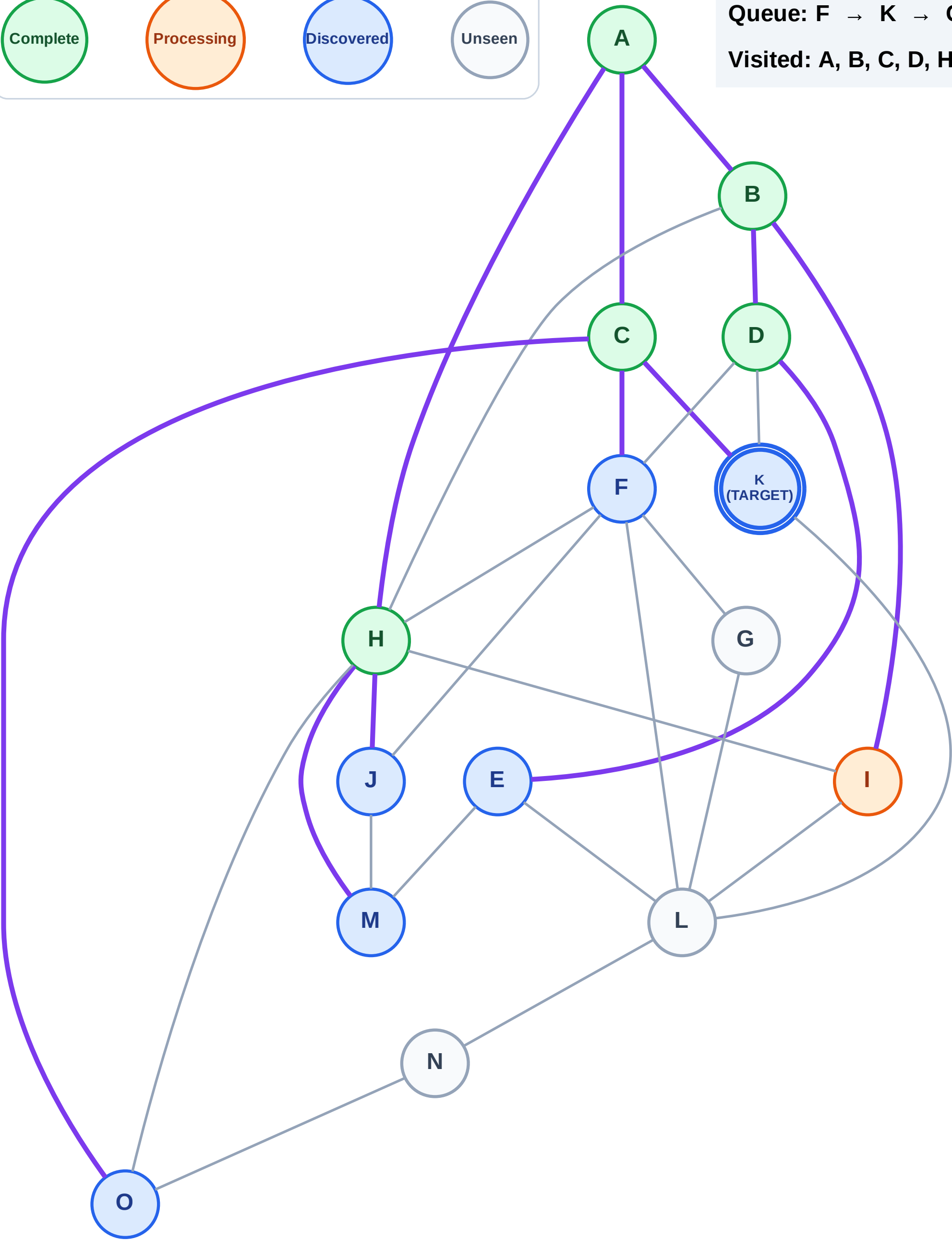
Processing

Discovered

Unseen

Queue: F → K → O → J → M → E

Visited: A, B, C, D, H



# BFS Traversal

Step 13: Discover L from I

## Legend

Complete

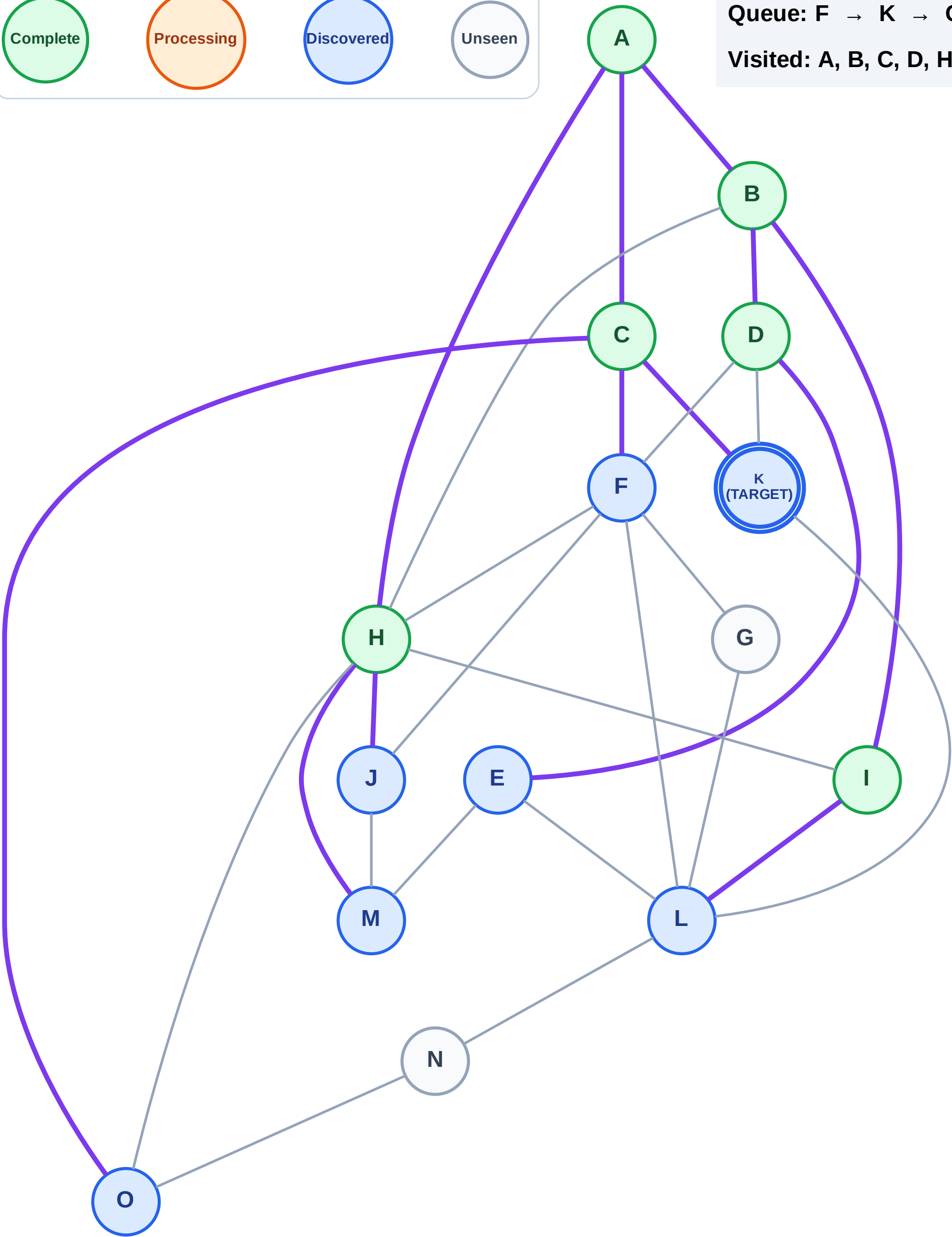
Processing

Discovered

Unseen

Queue: F → K → O → J → M → E → L

Visited: A, B, C, D, H, I



# BFS Traversal

Step 14: Process node F

Legend

Complete

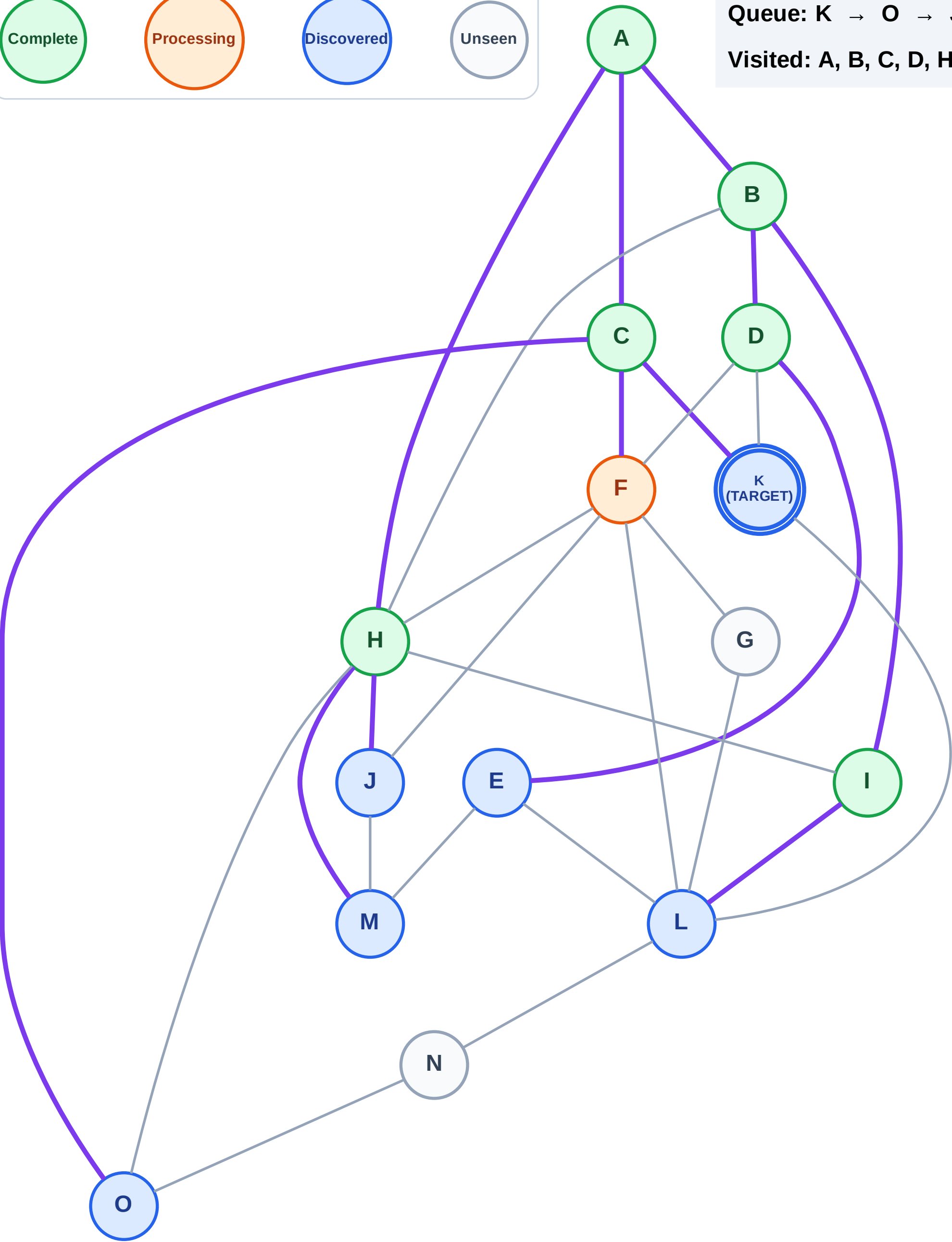
Processing

Discovered

Unseen

Queue: K → O → J → M → E → L

Visited: A, B, C, D, H, I



# BFS Traversal

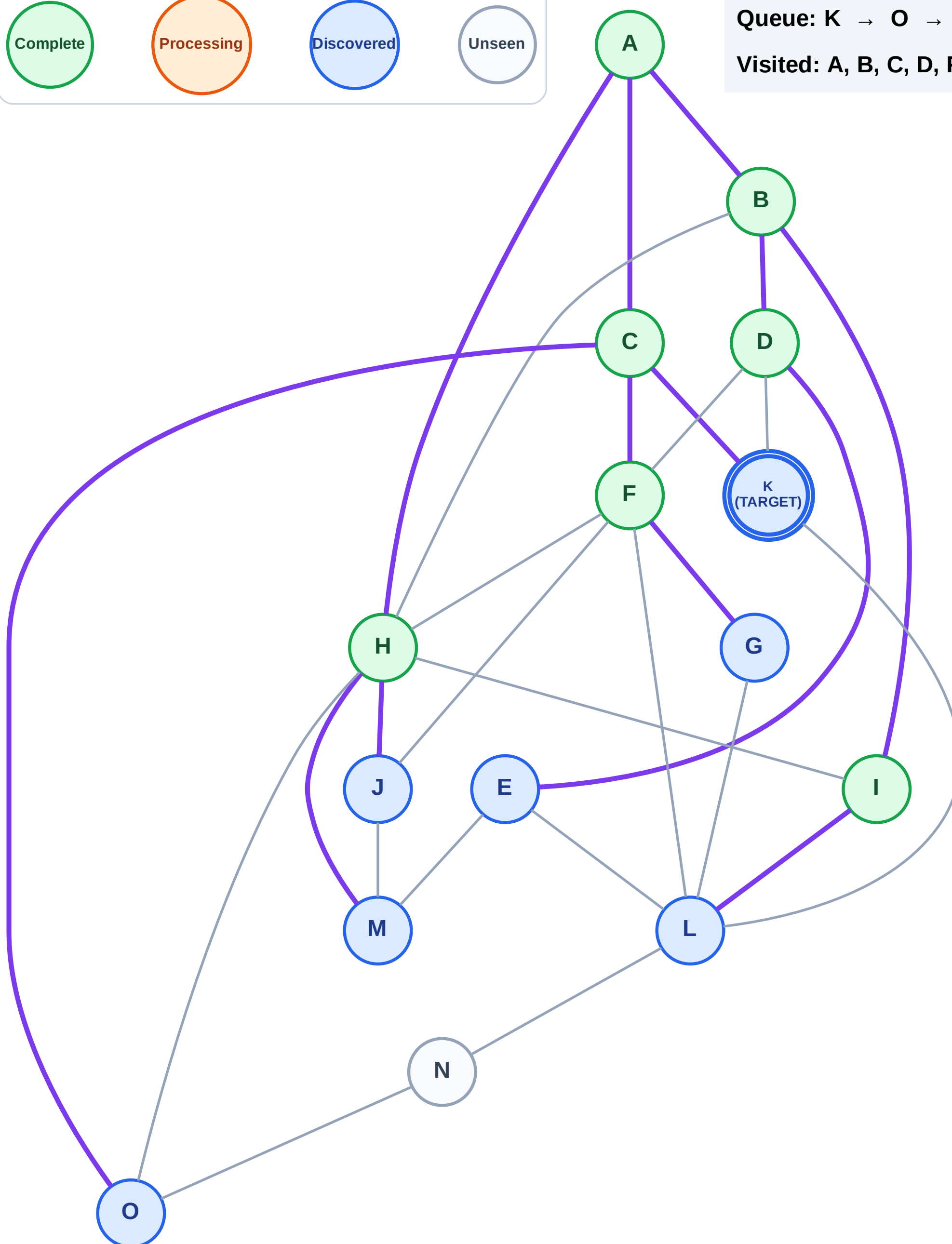
Step 15: Discover G from F

## Legend



Queue: K → O → J → M → E → L → G

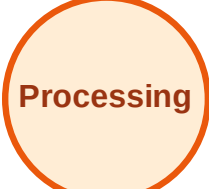
Visited: A, B, C, D, F, H, I



# BFS Traversal

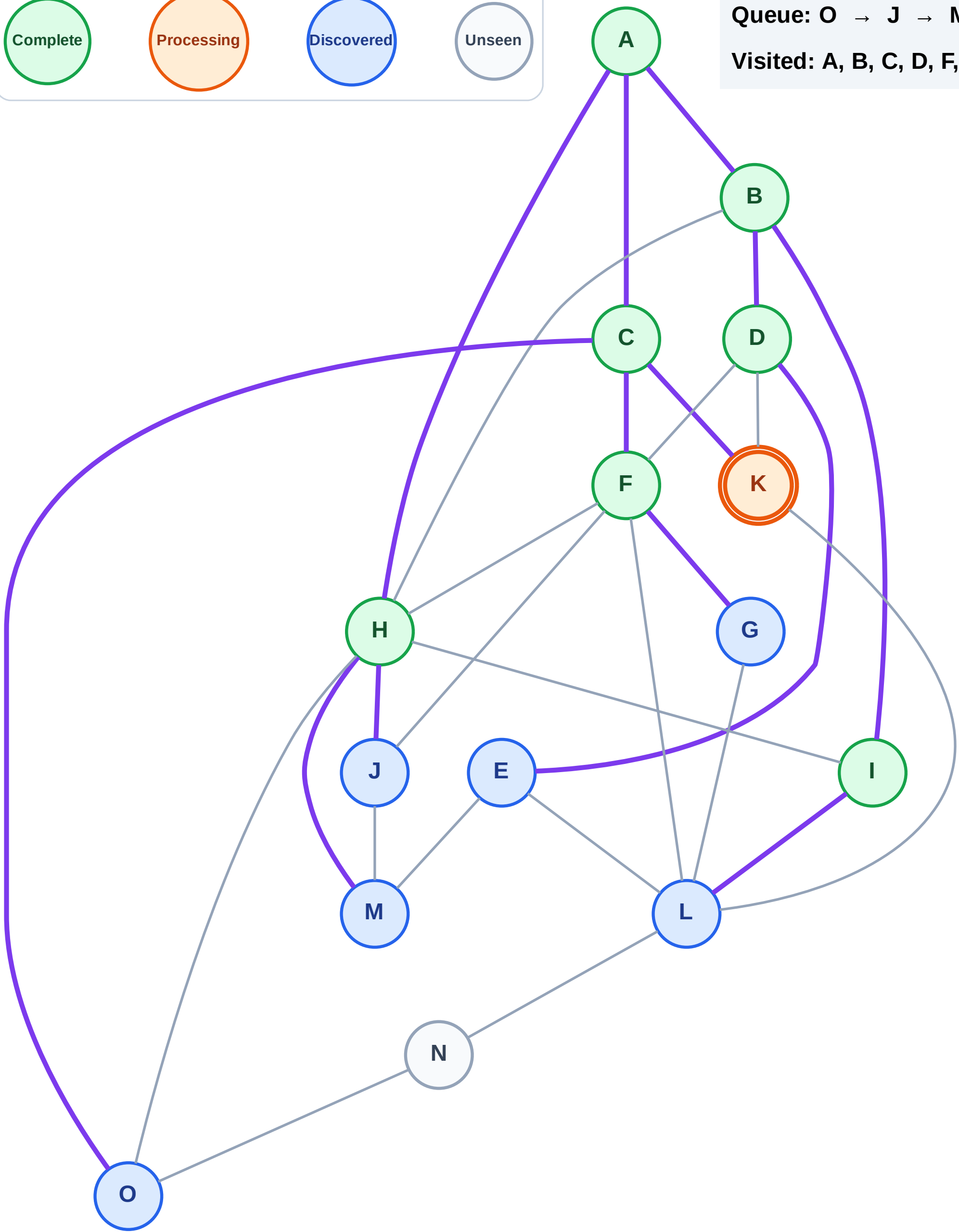
Step 16: Process node K

## Legend



Queue: O → J → M → E → L → G

Visited: A, B, C, D, F, H, I



# BFS Traversal

Step 17: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend

Complete

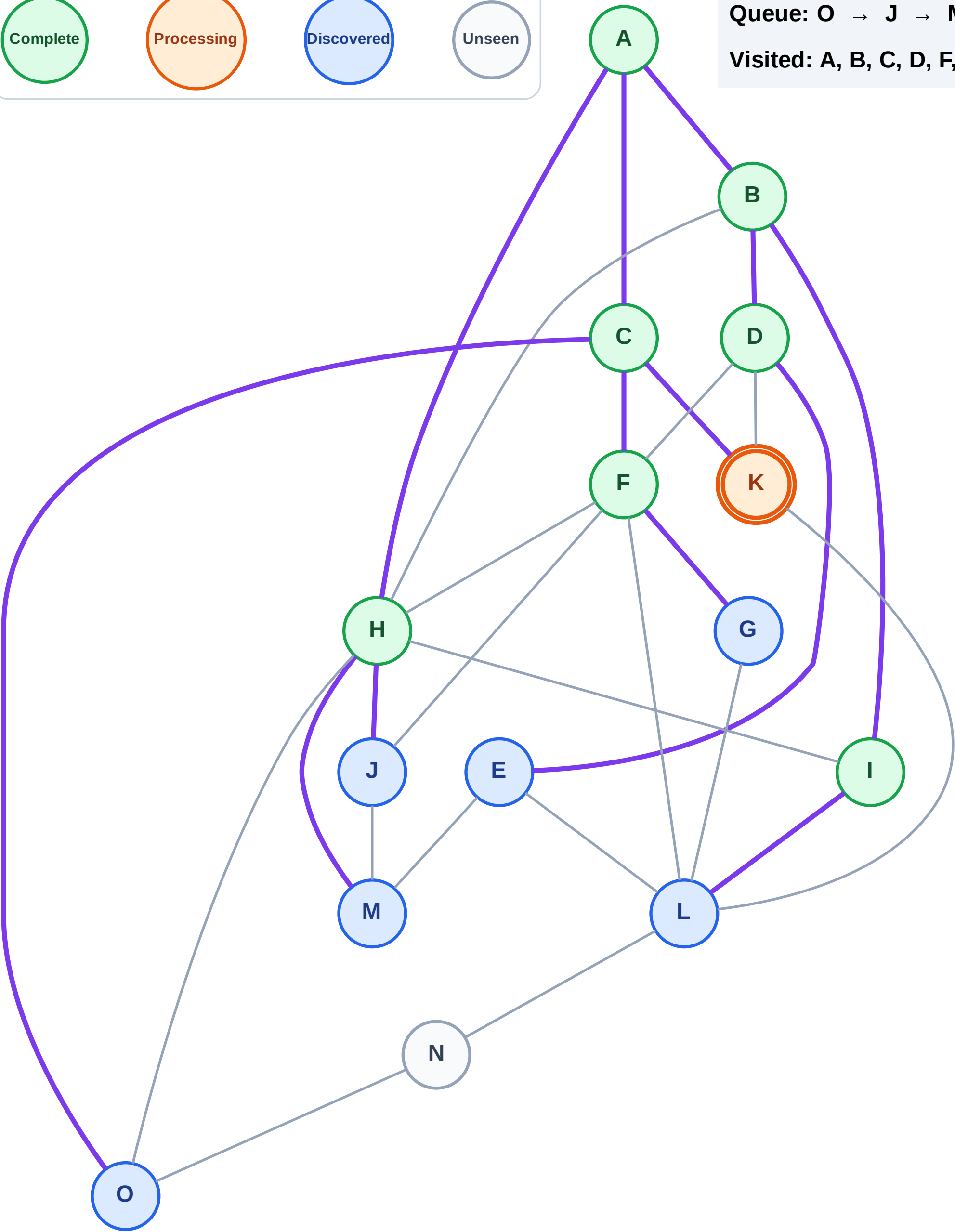
Processing

Discovered

Unseen

Queue: O → J → M → E → L → G

Visited: A, B, C, D, F, H, I, K



# DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

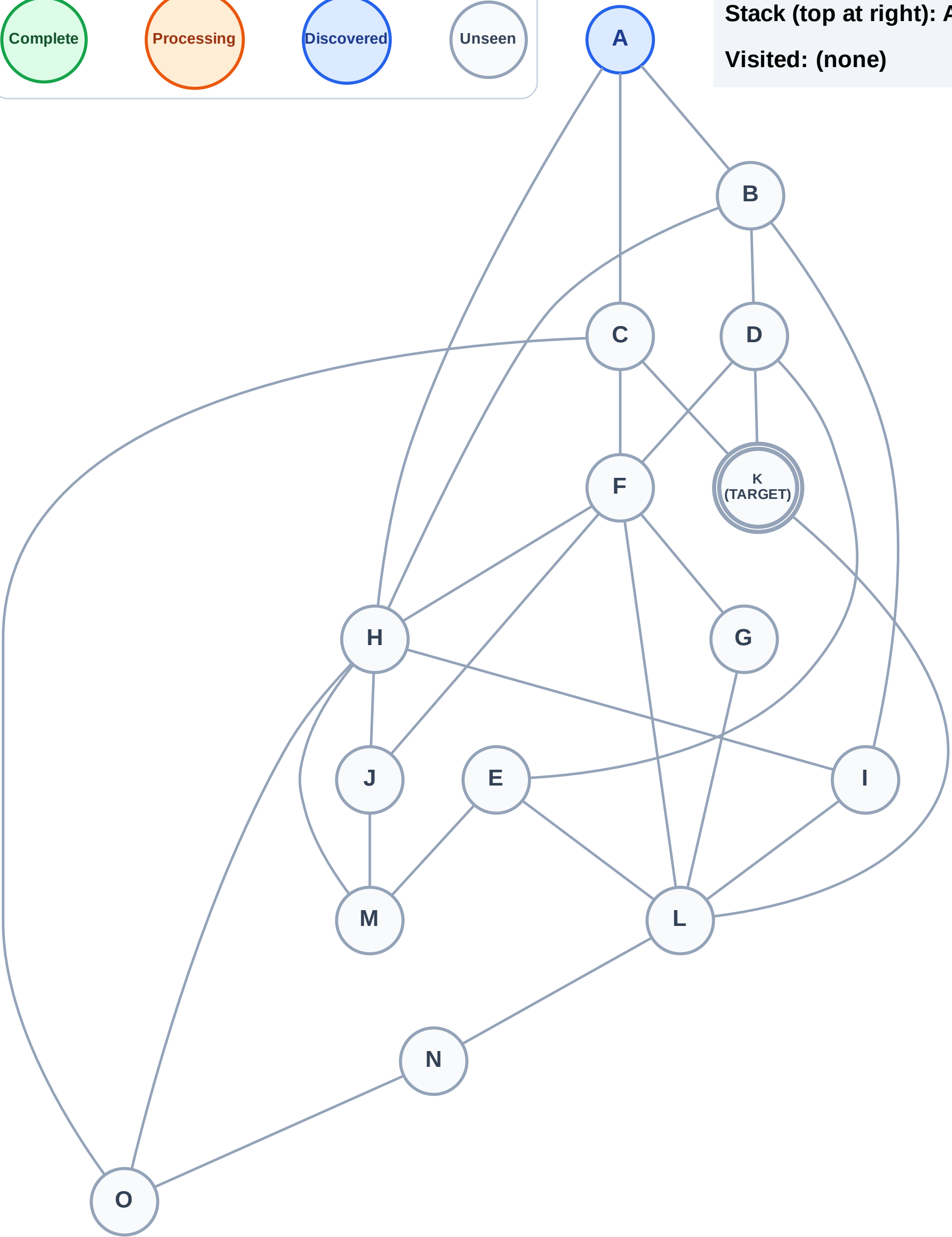
Complete

Processing

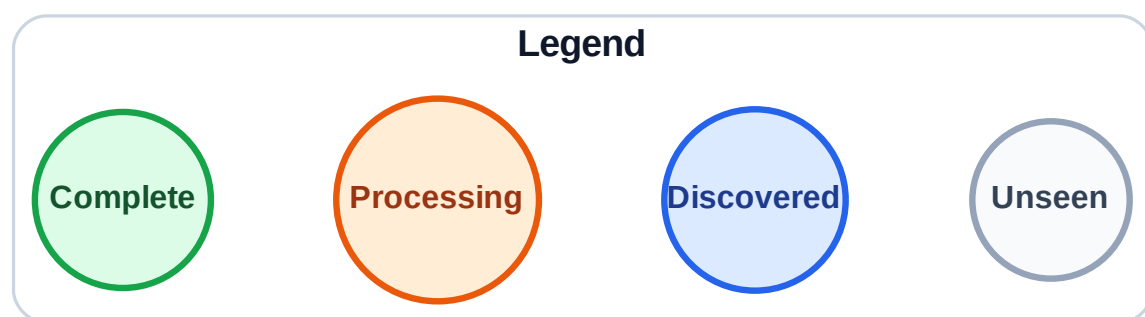
Discovered

Unseen

Stack (top at right): A  
Visited: (none)

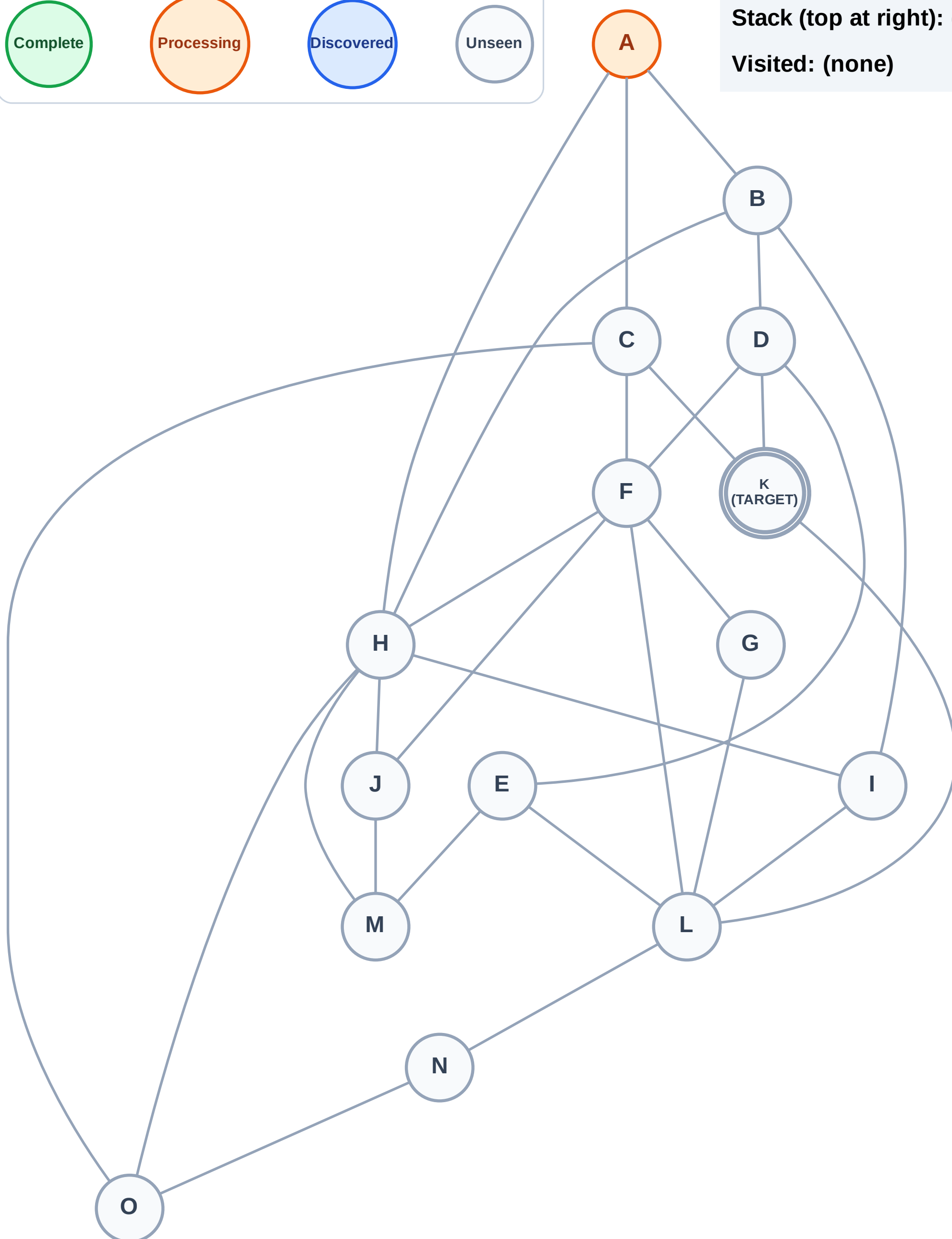


Step 2: Process node A



**Stack (top at right): (empty)**

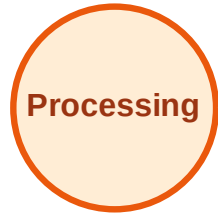
**Visited: (none)**



# DFS Traversal

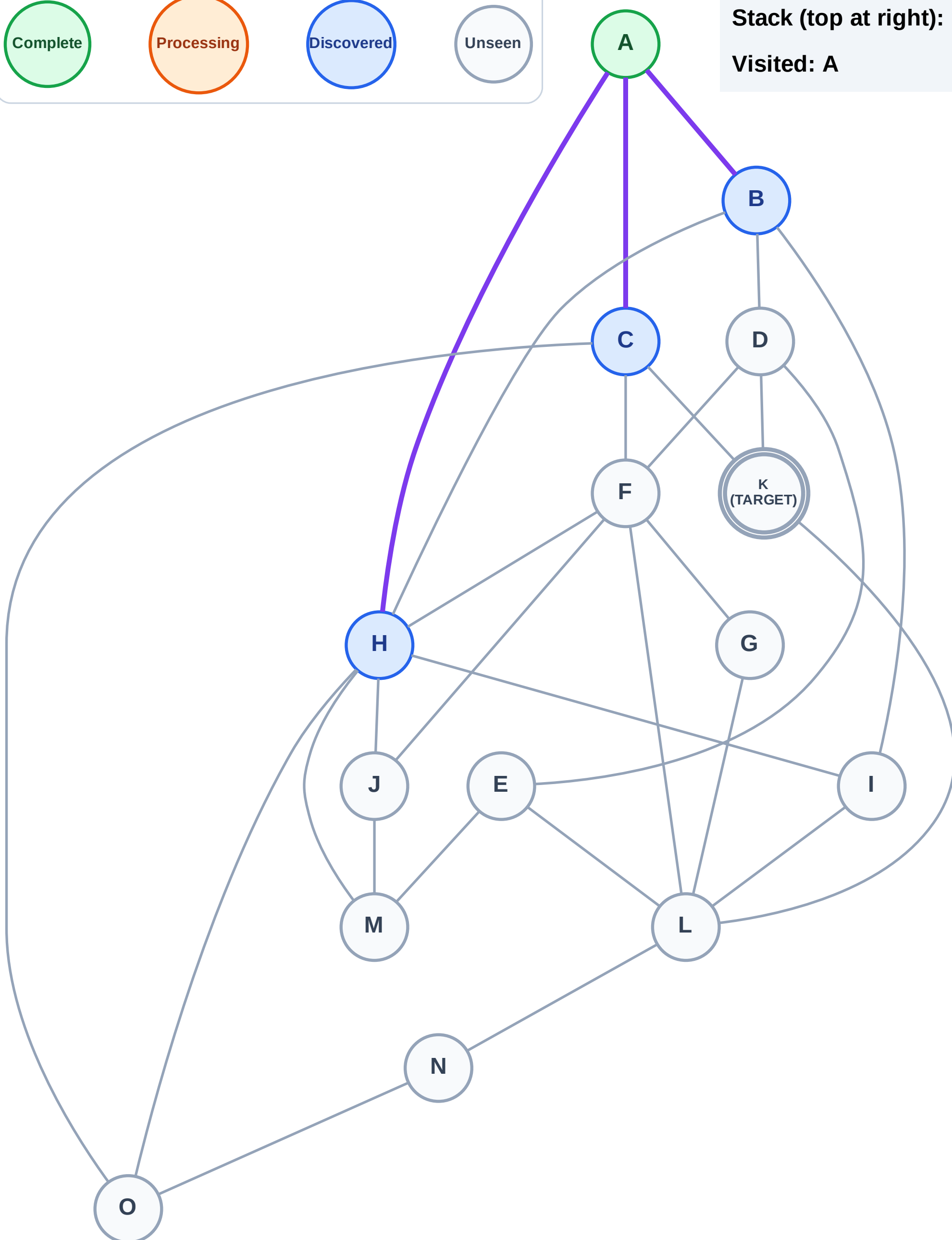
Step 3: Discover H, C, B from A

## Legend



Stack (top at right): H | C | B

Visited: A



# DFS Traversal

Step 4: Process node B

Legend

Complete

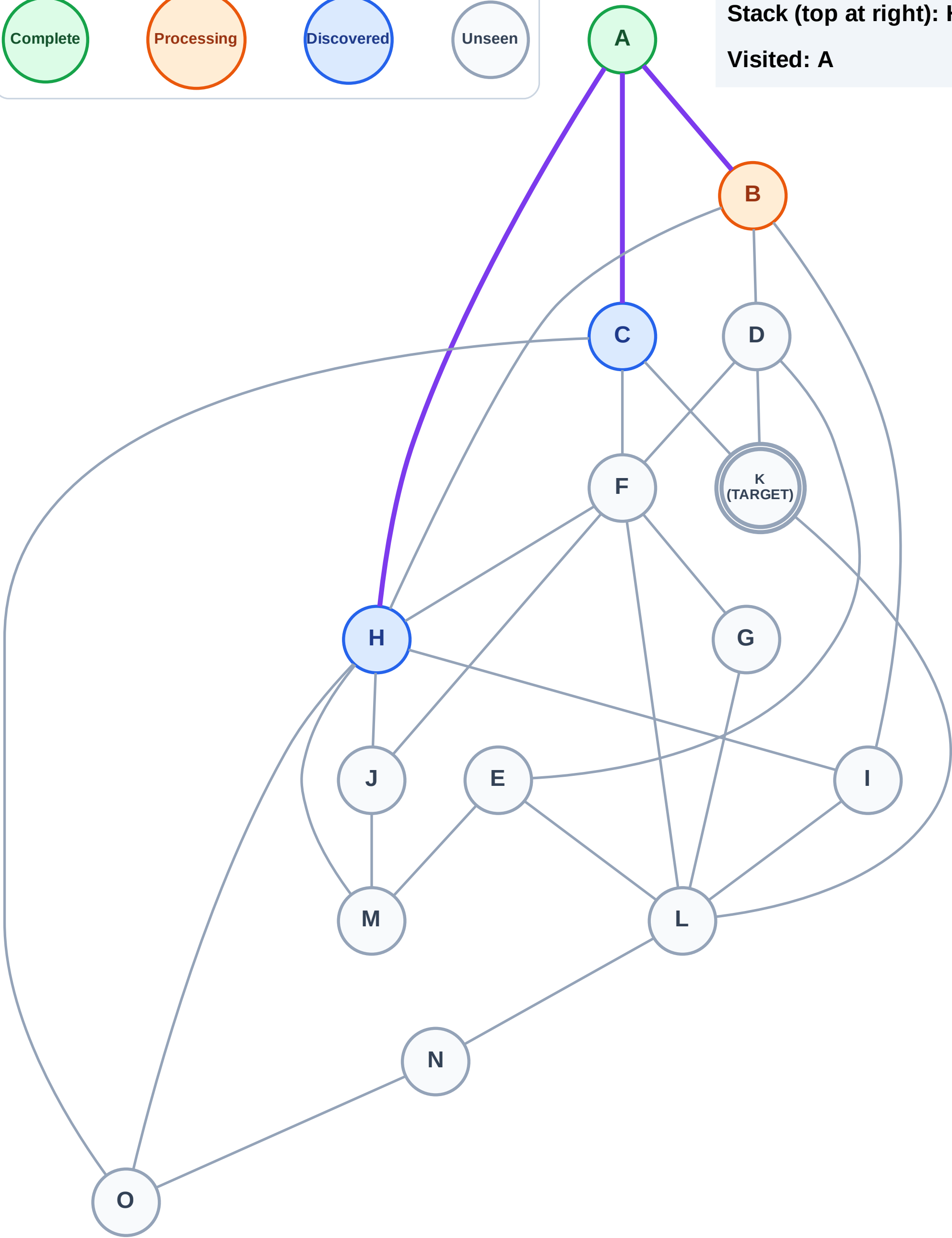
Processing

Discovered

Unseen

Stack (top at right): H | C

Visited: A



# DFS Traversal

Step 5: Discover I, D from B

Legend

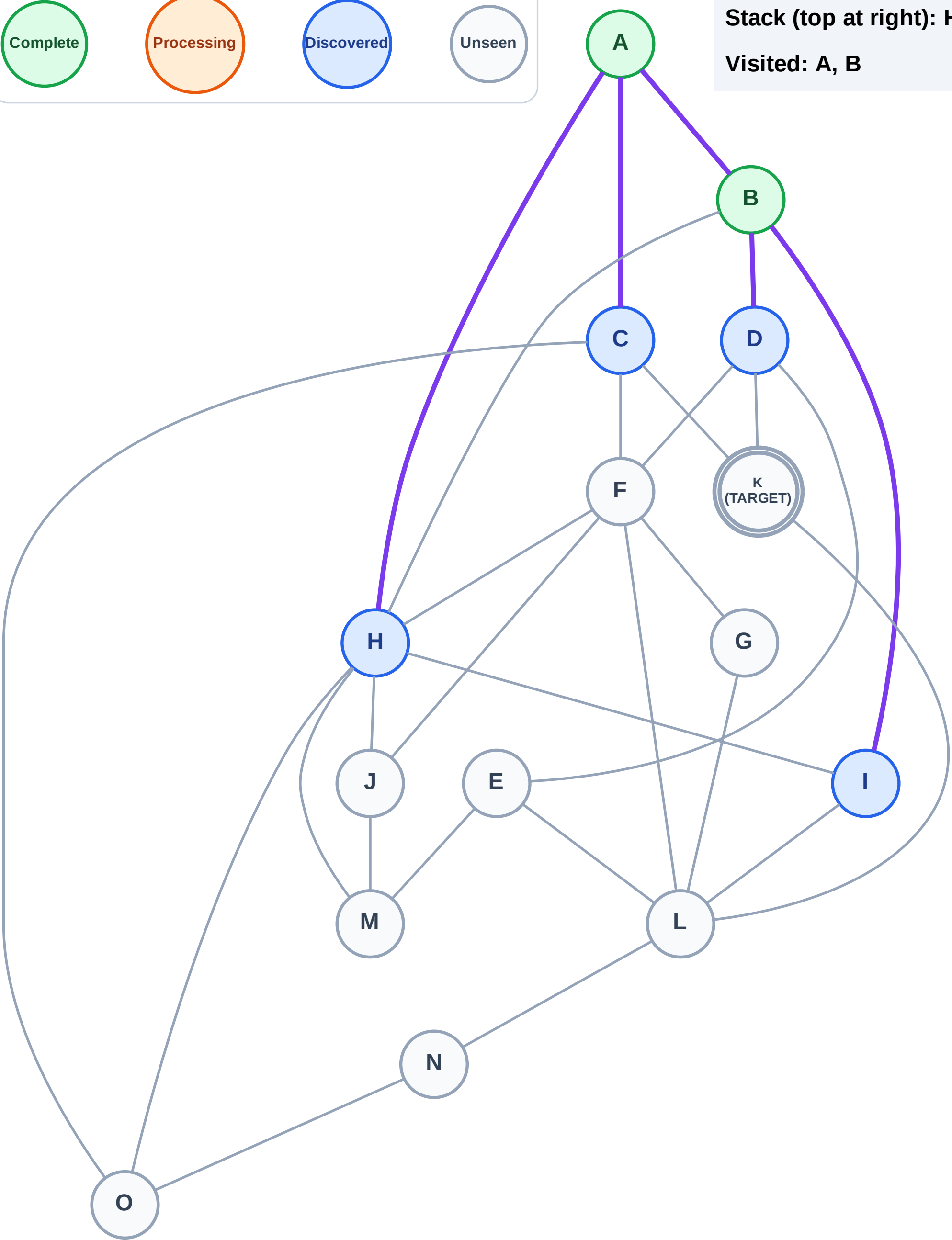
Complete

Processing

Discovered

Unseen

Stack (top at right): H | C | I | D  
Visited: A, B



# DFS Traversal

Step 6: Process node D

Legend

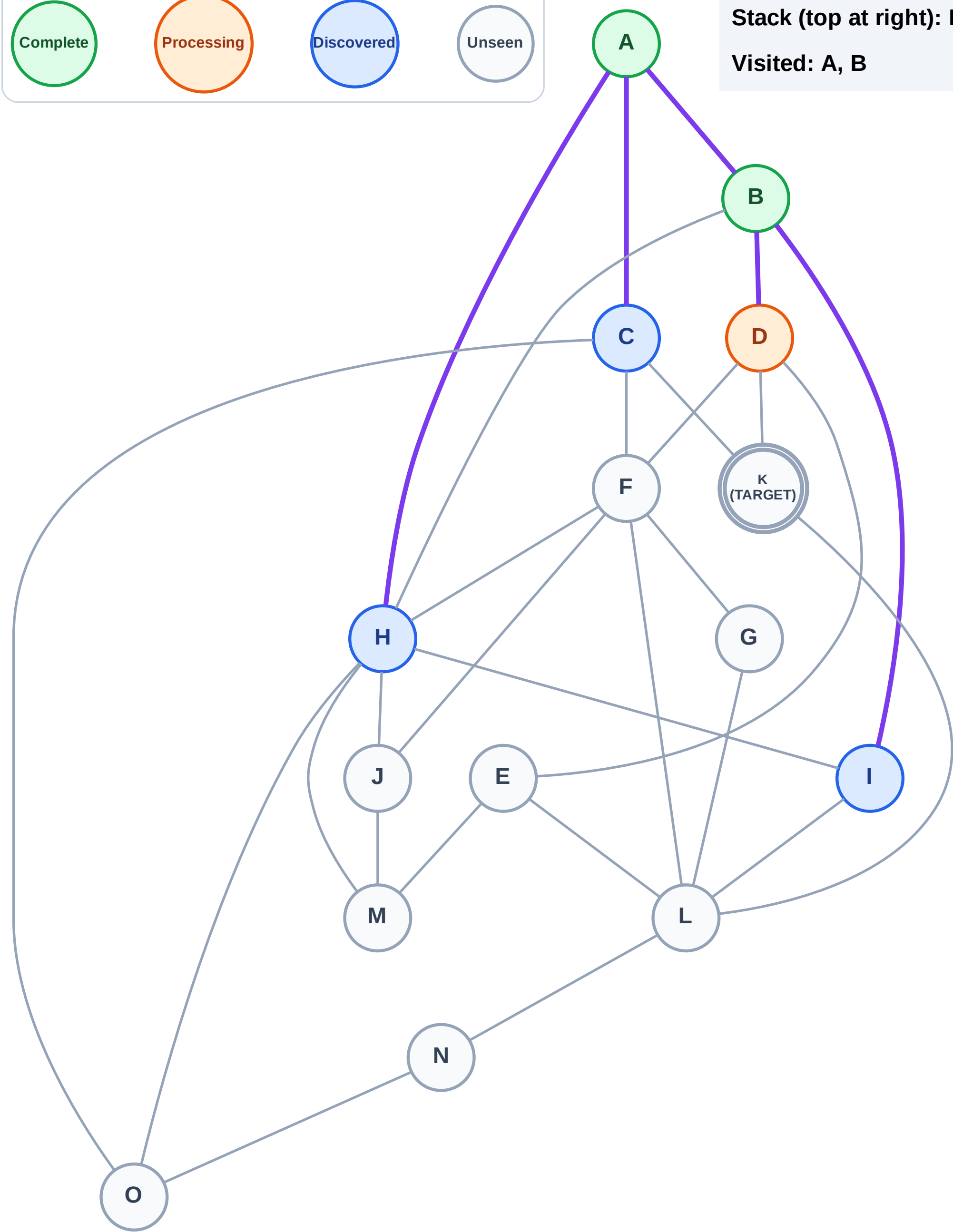
Complete

Processing

Discovered

Unseen

Stack (top at right): H | C | I  
Visited: A, B



# DFS Traversal

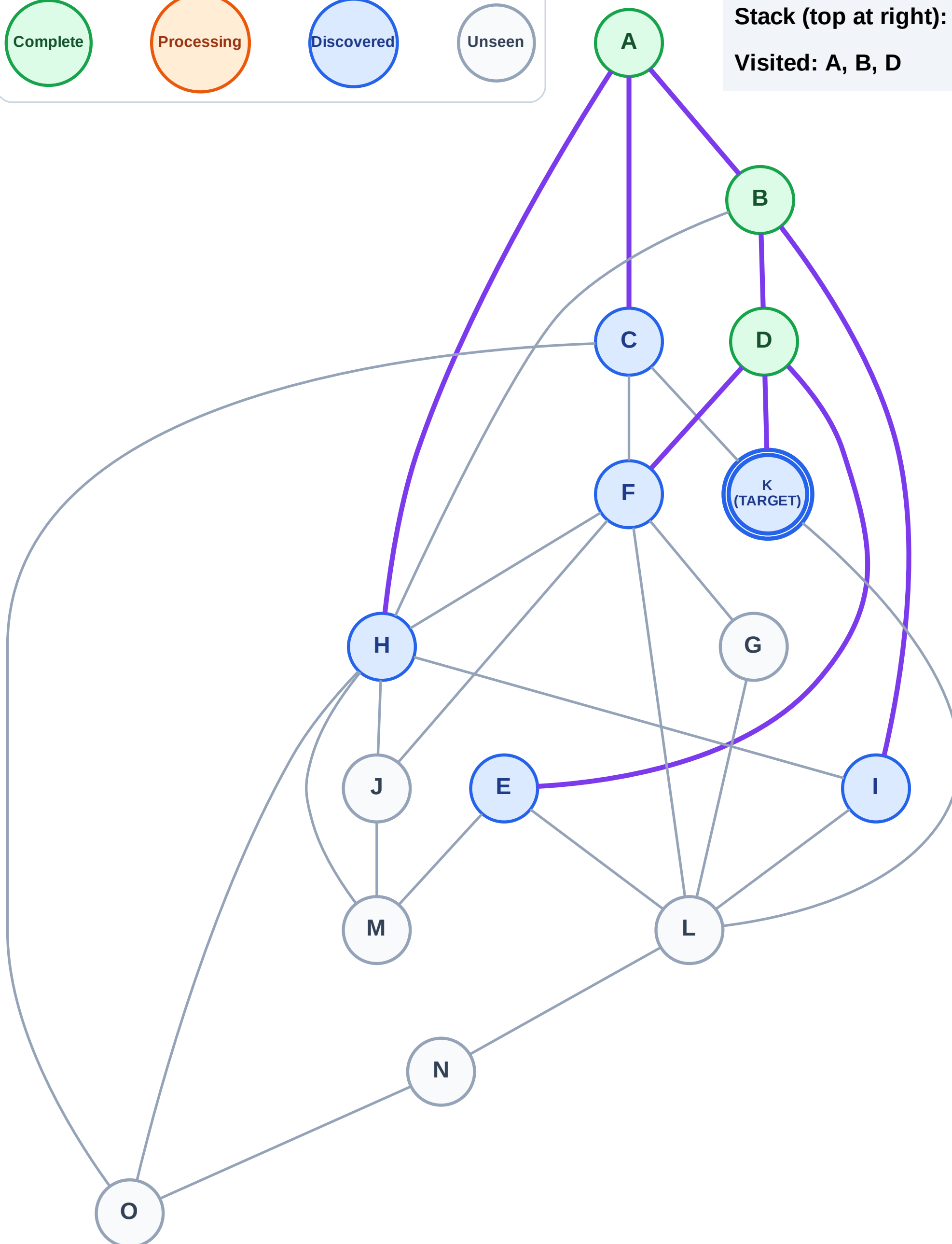
Step 7: Discover K, F, E from D

## Legend



Stack (top at right): H | C | I | K | F | E

Visited: A, B, D



# DFS Traversal

Step 8: Process node E

Legend

Complete

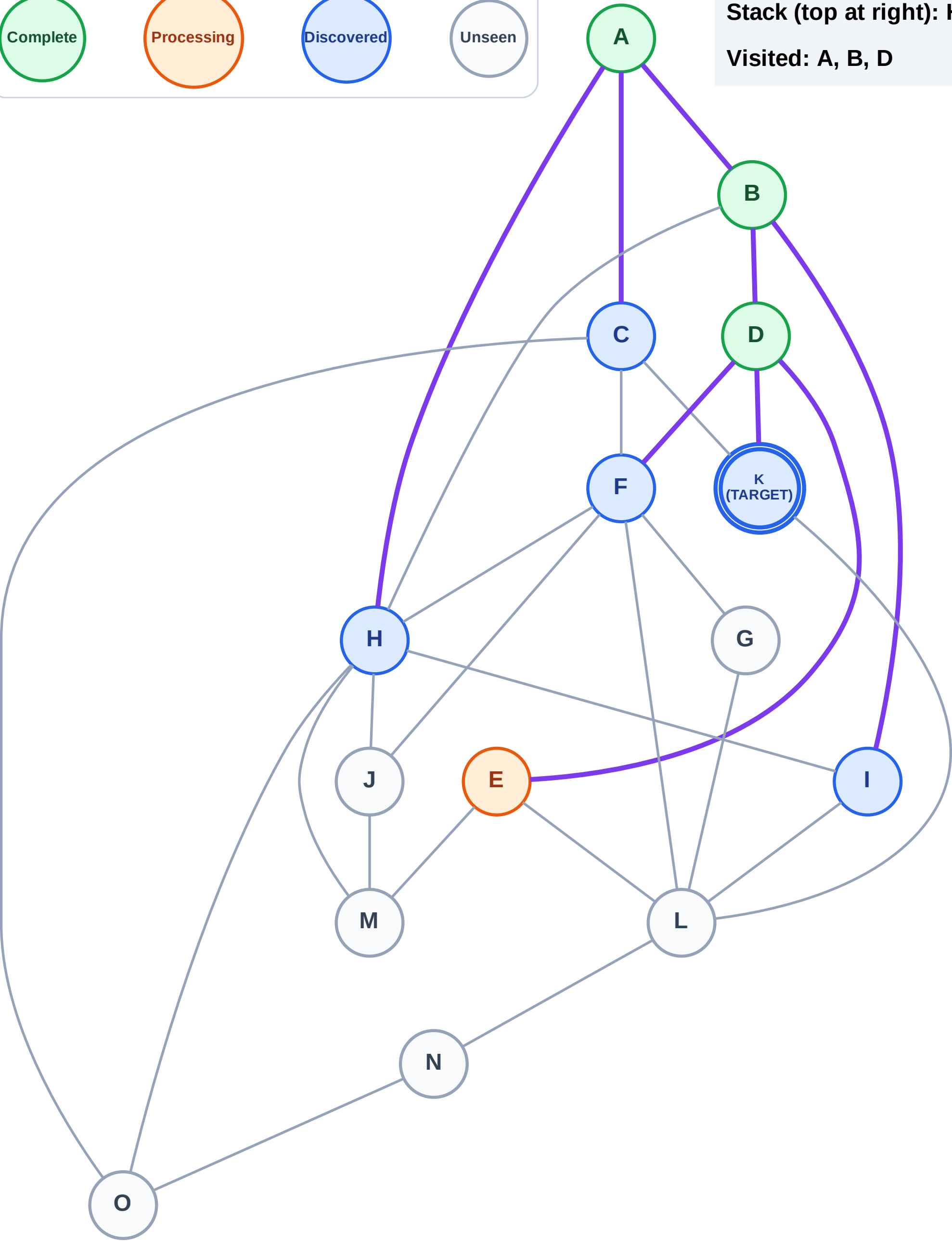
Processing

Discovered

Unseen

Stack (top at right): H | C | I | K | F

Visited: A, B, D



# DFS Traversal

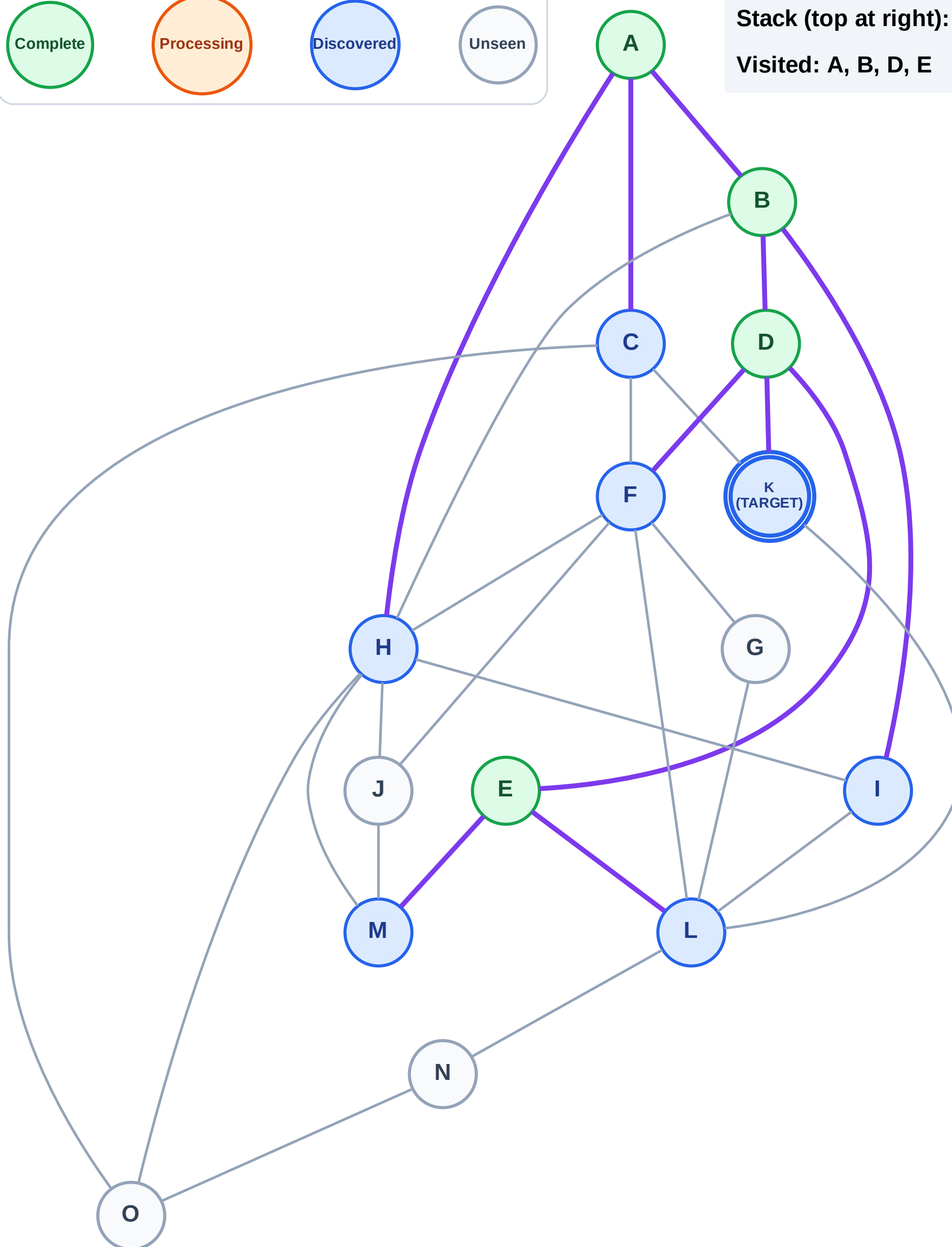
Step 9: Discover M, L from E

## Legend



Stack (top at right): H | C | I | K | F | M | L

Visited: A, B, D, E



# DFS Traversal

Step 10: Process node L

## Legend

Complete

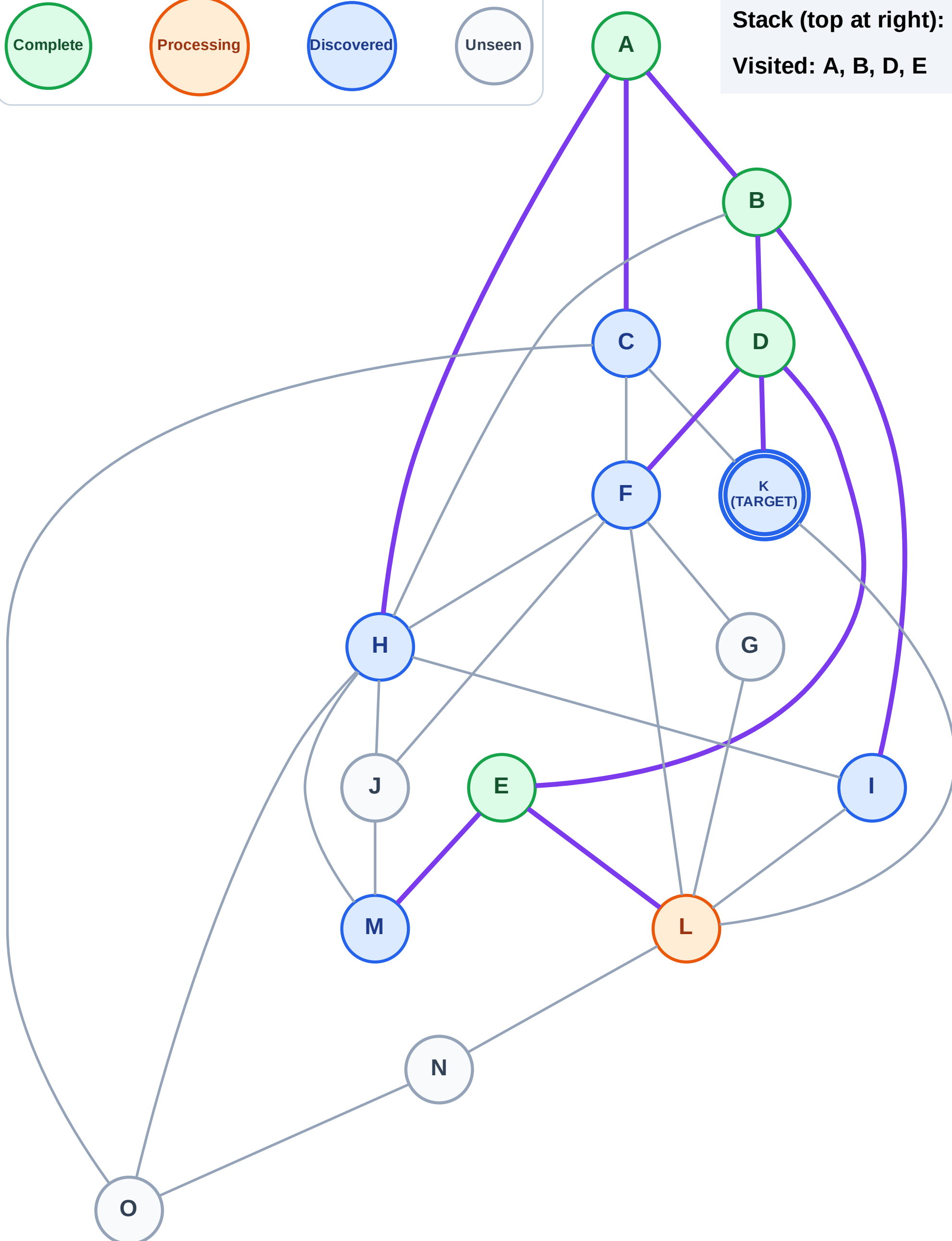
Processing

Discovered

Unseen

Stack (top at right): H | C | I | K | F | M

Visited: A, B, D, E



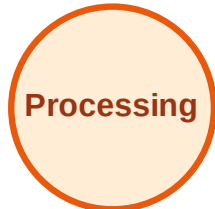
# DFS Traversal

Step 11: Discover N, G from L

## Legend



Complete



Processing



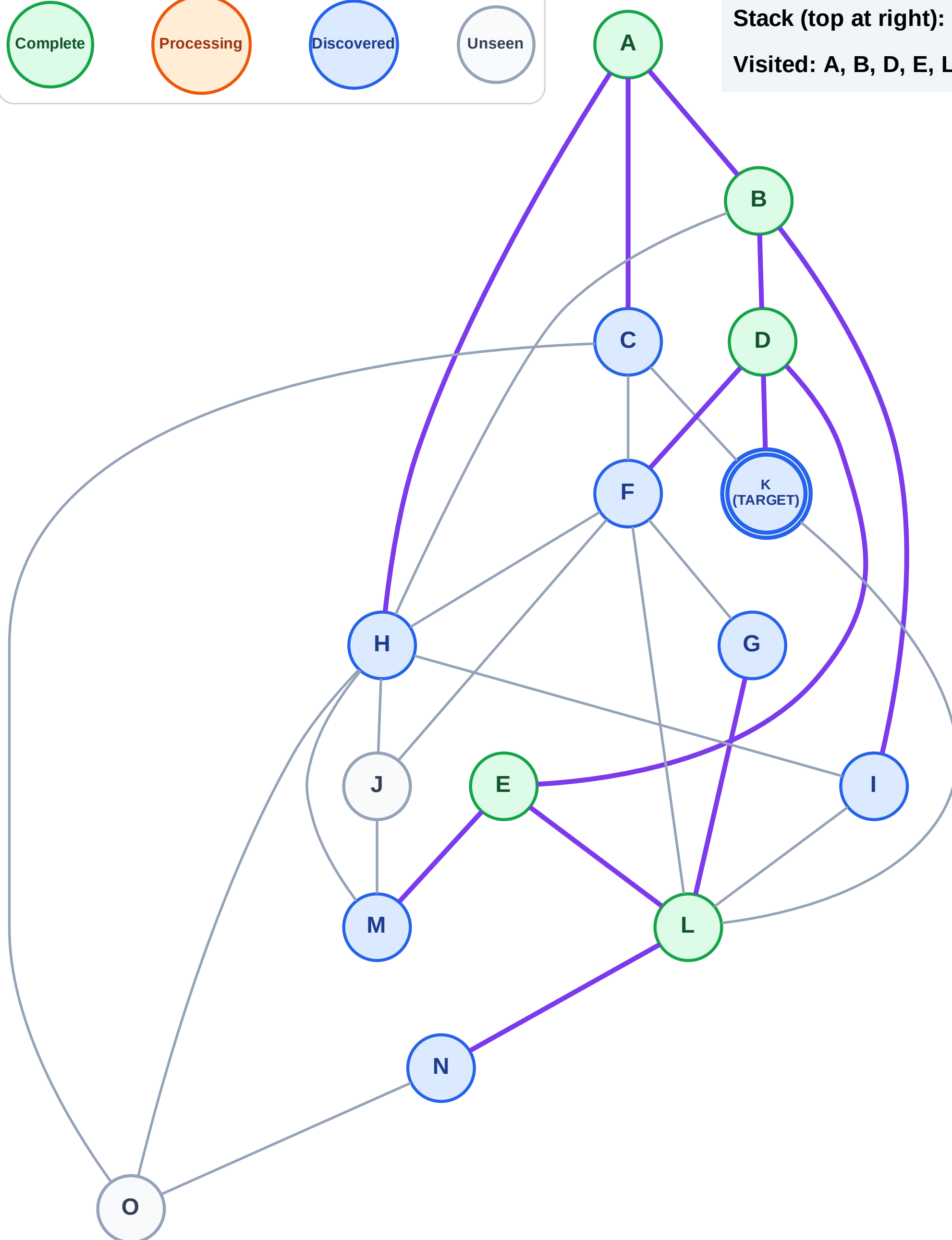
Discovered



Unseen

Stack (top at right): H | C | I | K | F | M | N | G

Visited: A, B, D, E, L



# DFS Traversal

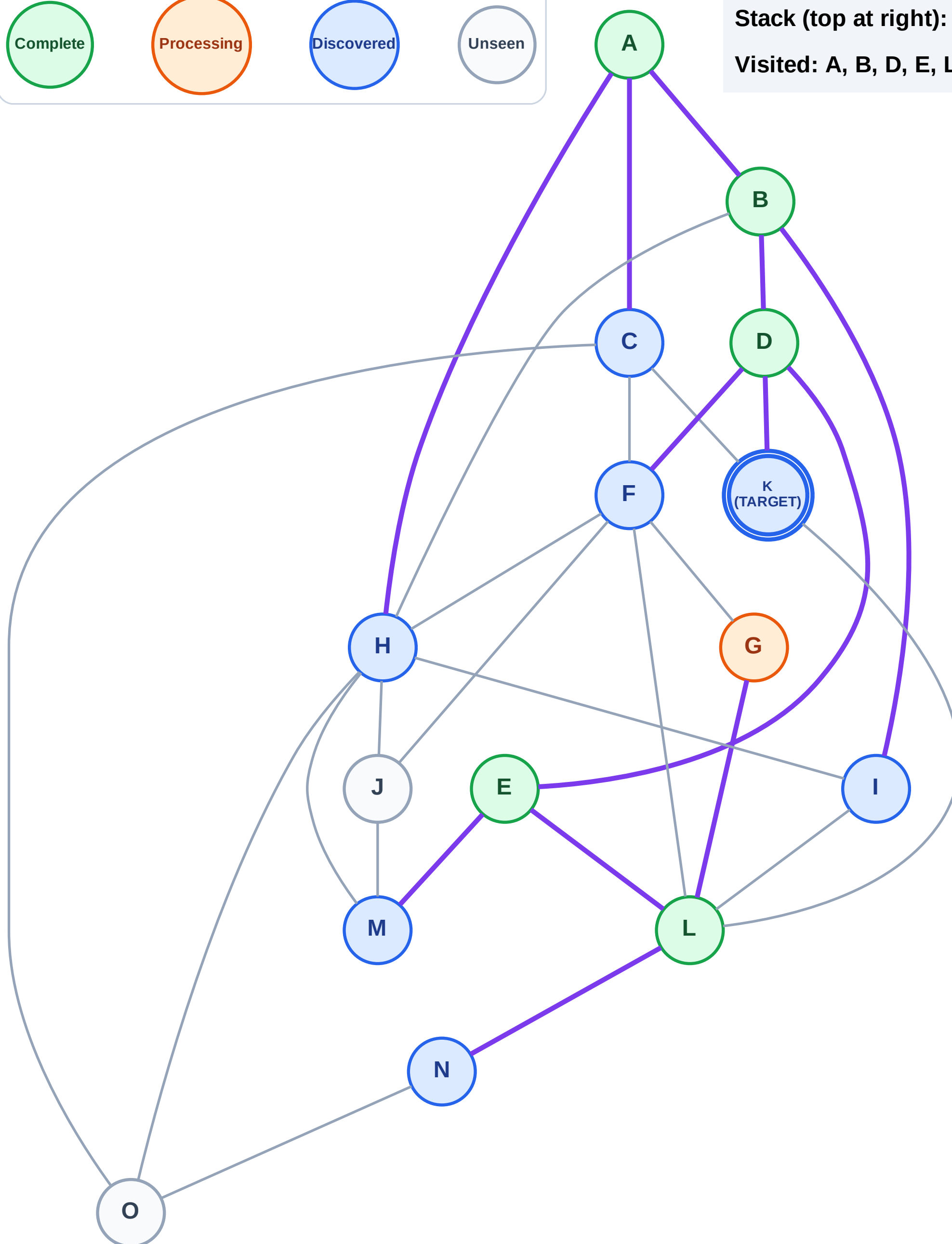
Step 12: Process node G

## Legend



Stack (top at right): H | C | I | K | F | M | N

Visited: A, B, D, E, L



# DFS Traversal

Step 13: No new neighbors from G

Legend

Complete

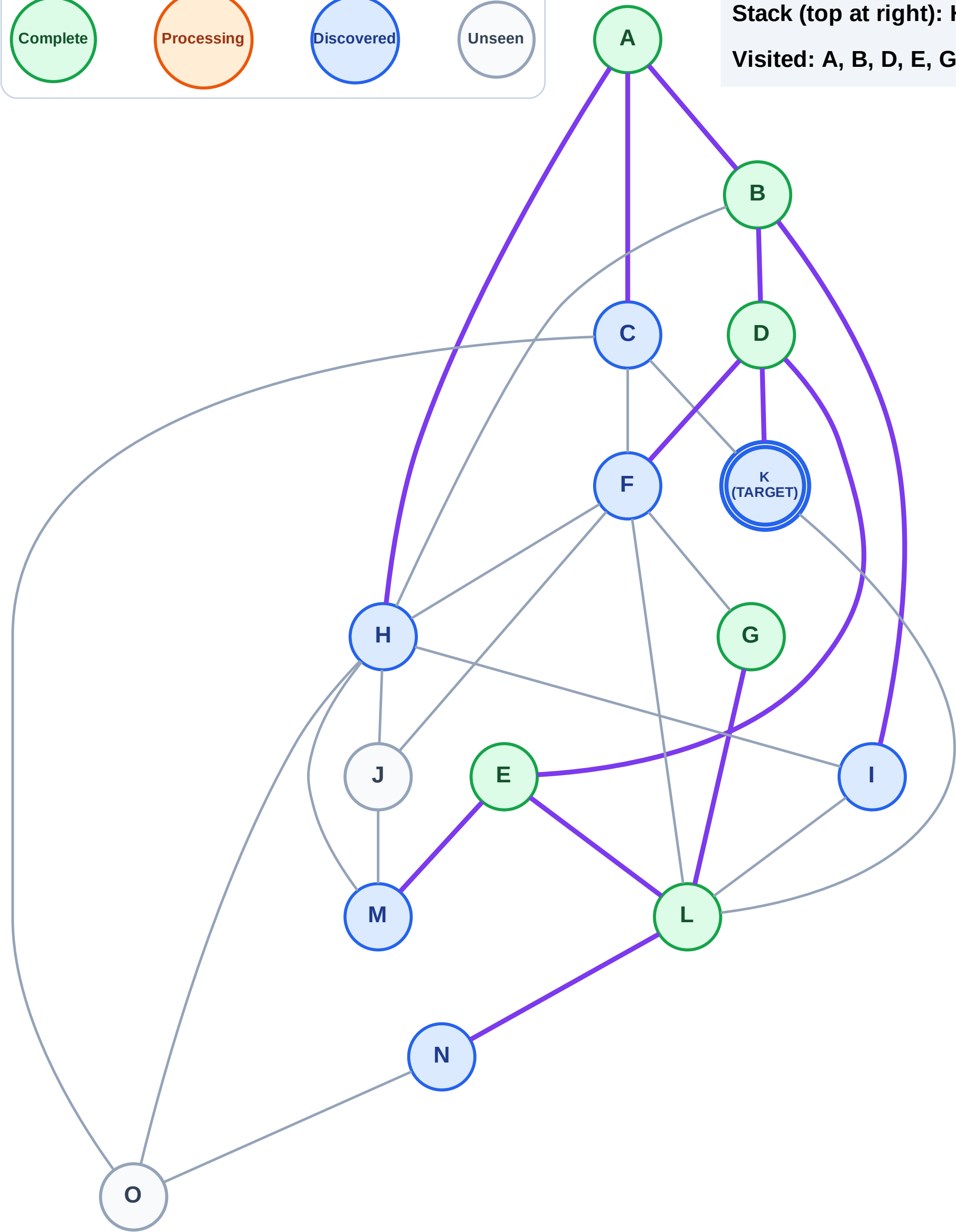
Processing

Discovered

Unseen

Stack (top at right): H | C | I | K | F | M | N

Visited: A, B, D, E, G, L



# DFS Traversal

Step 14: Process node N

## Legend

Complete

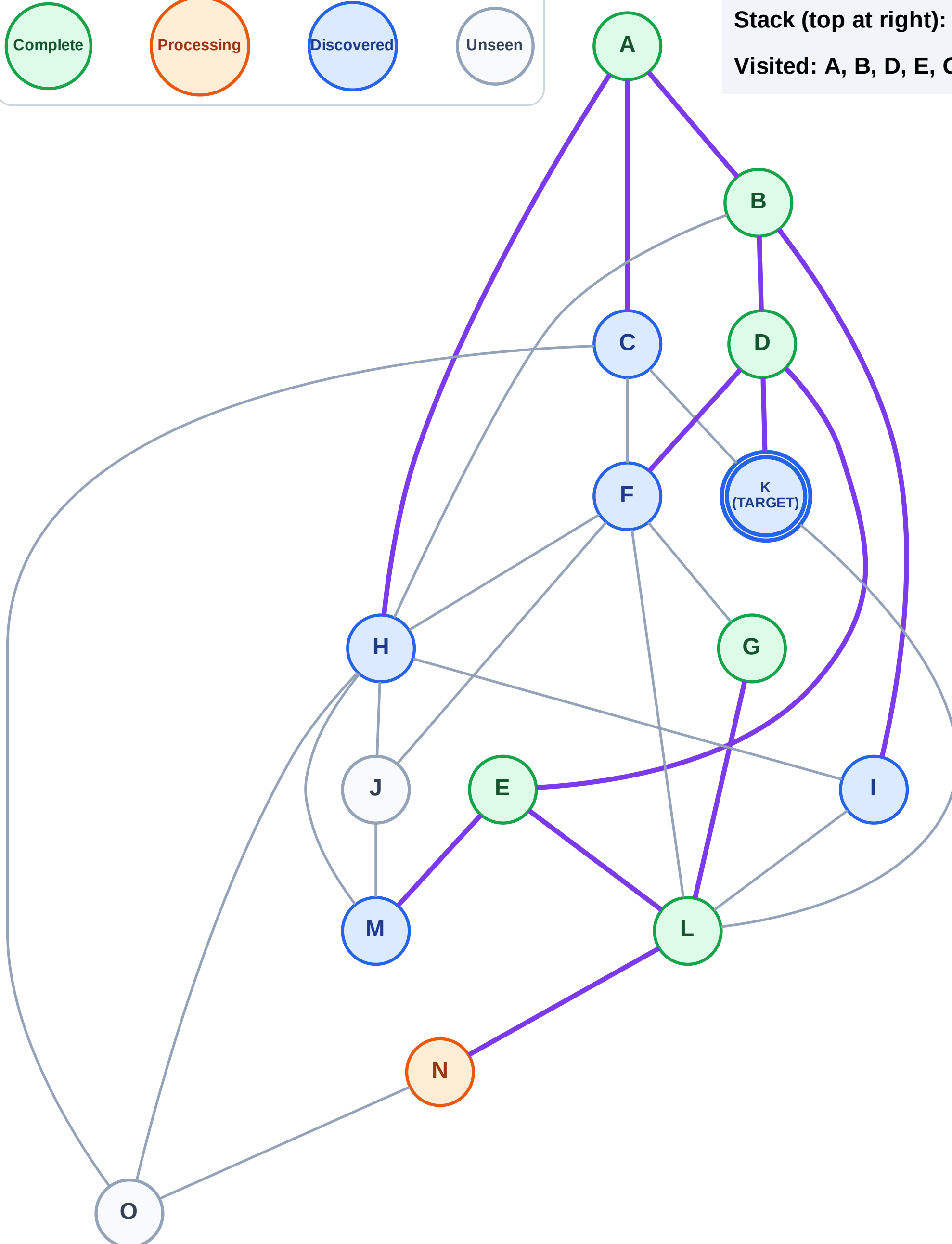
Processing

Discovered

Unseen

Stack (top at right): H | C | I | K | F | M

Visited: A, B, D, E, G, L



# DFS Traversal

Step 15: Discover O from N

## Legend



Complete



Processing



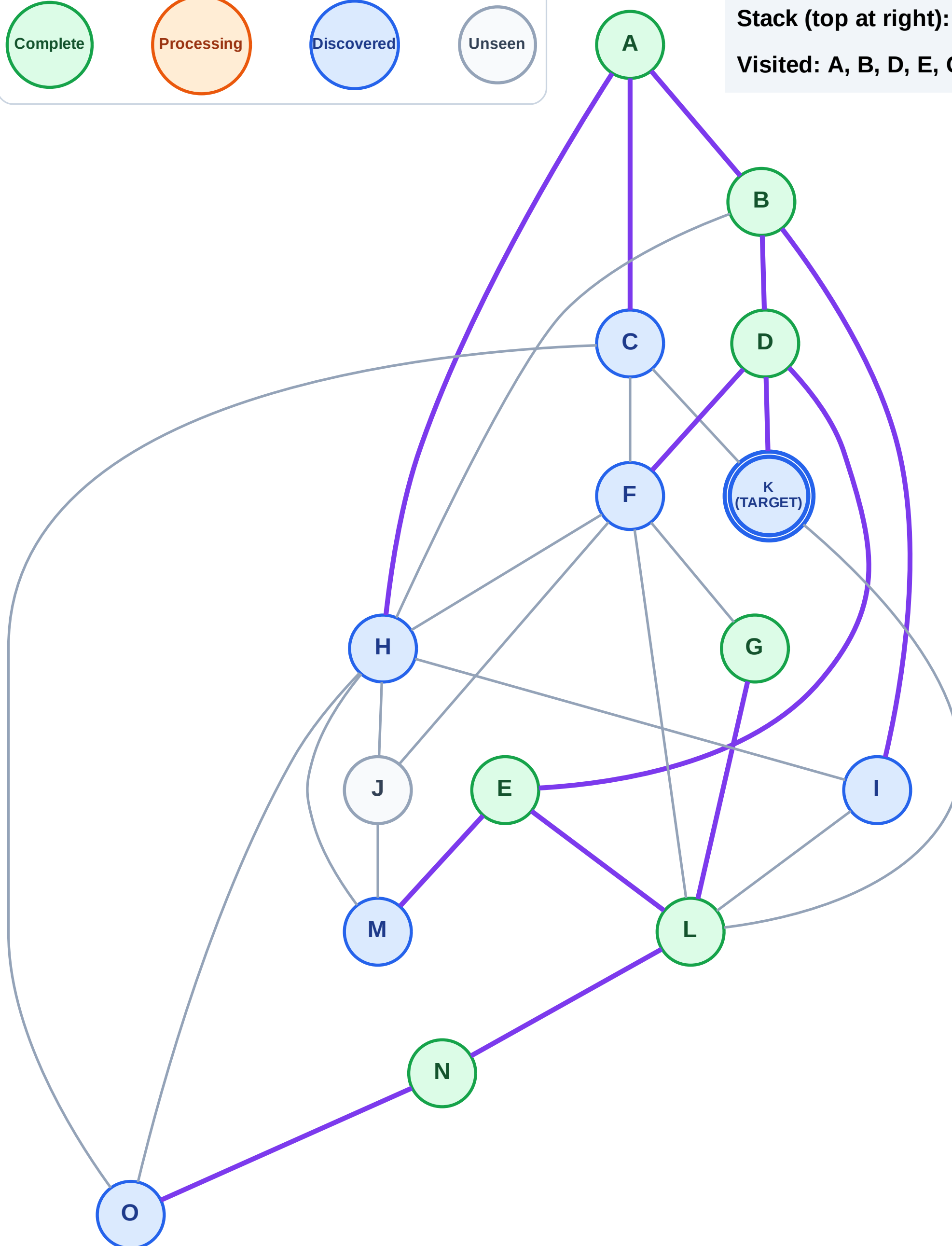
Discovered



Unseen

Stack (top at right): H | C | I | K | F | M | O

Visited: A, B, D, E, G, L, N



# DFS Traversal

Step 16: Process node O

## Legend

Complete

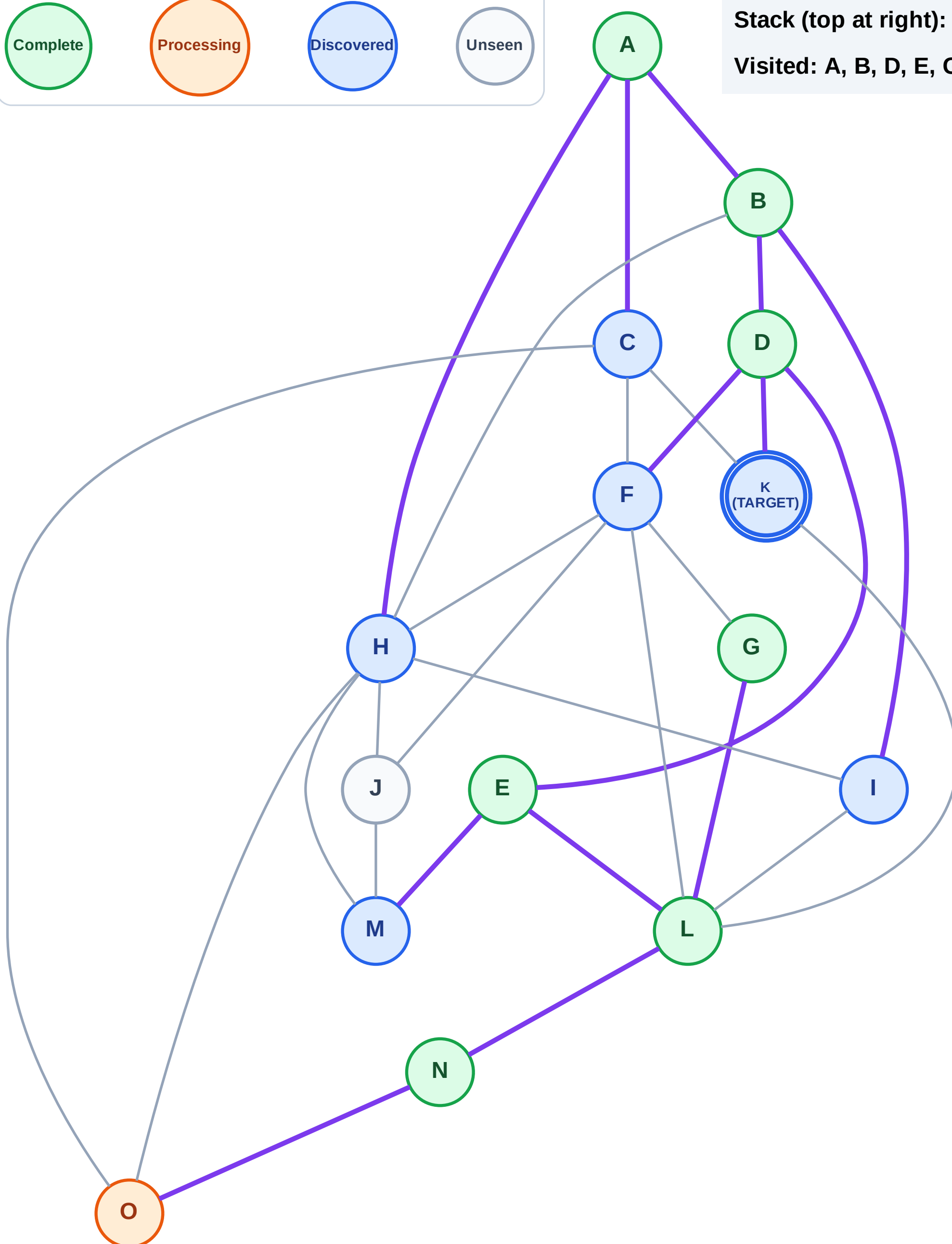
Processing

Discovered

Unseen

Stack (top at right): H | C | I | K | F | M

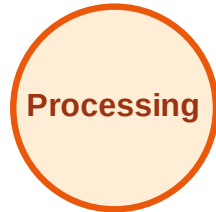
Visited: A, B, D, E, G, L, N



# DFS Traversal

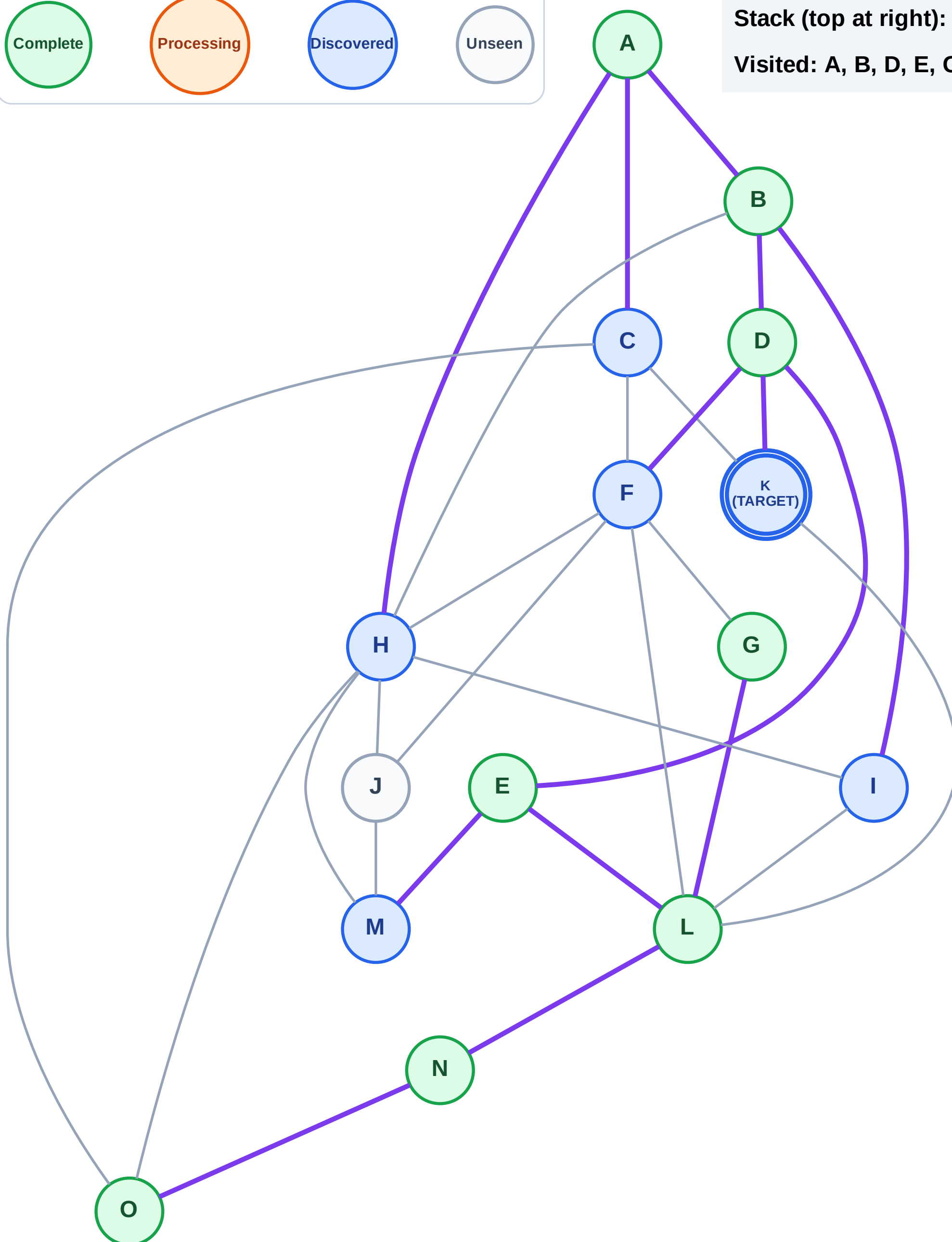
Step 17: No new neighbors from O

## Legend



Stack (top at right): H | C | I | K | F | M

Visited: A, B, D, E, G, L, N, O



# DFS Traversal

Step 18: Process node M

## Legend

Complete

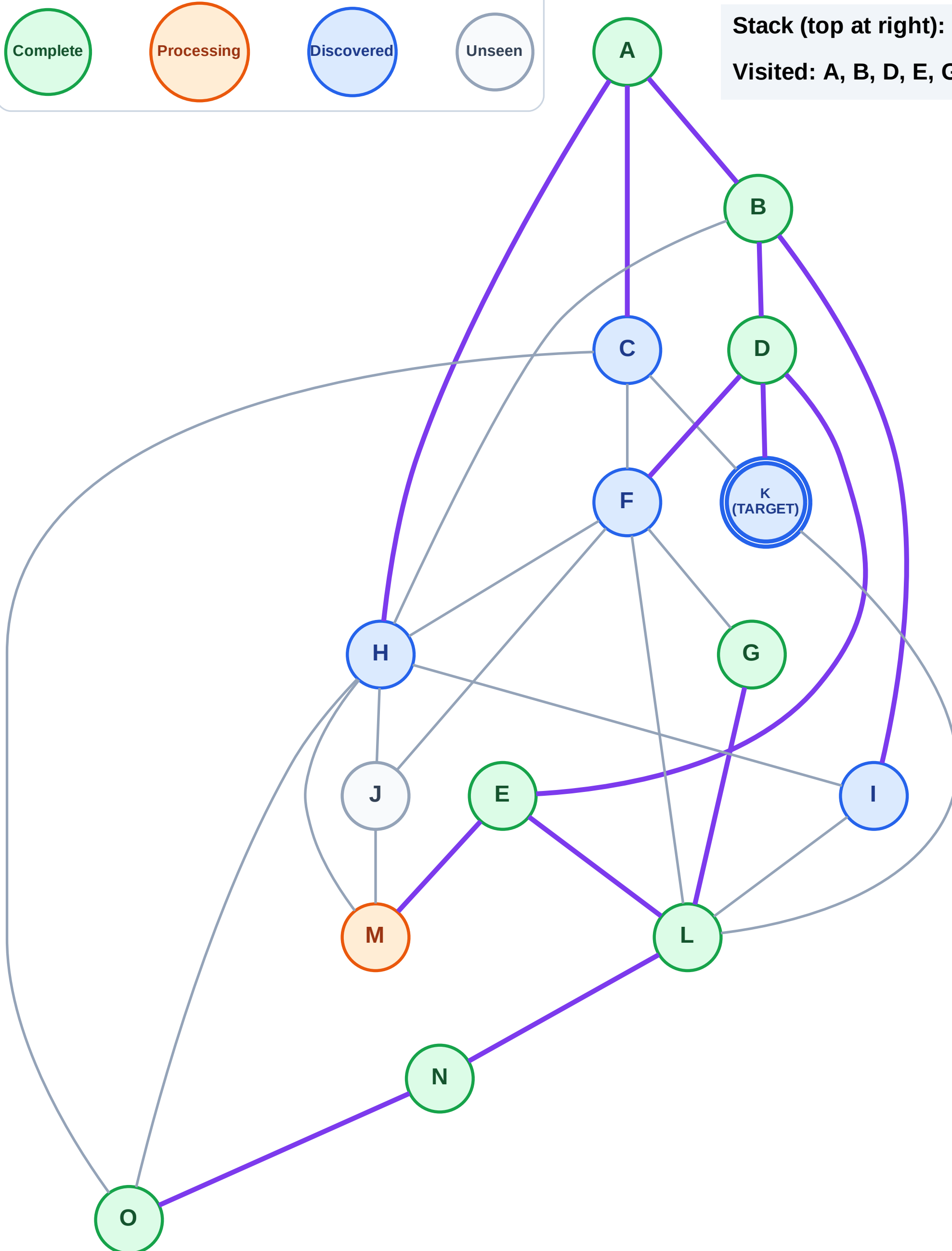
Processing

Discovered

Unseen

Stack (top at right): H | C | I | K | F

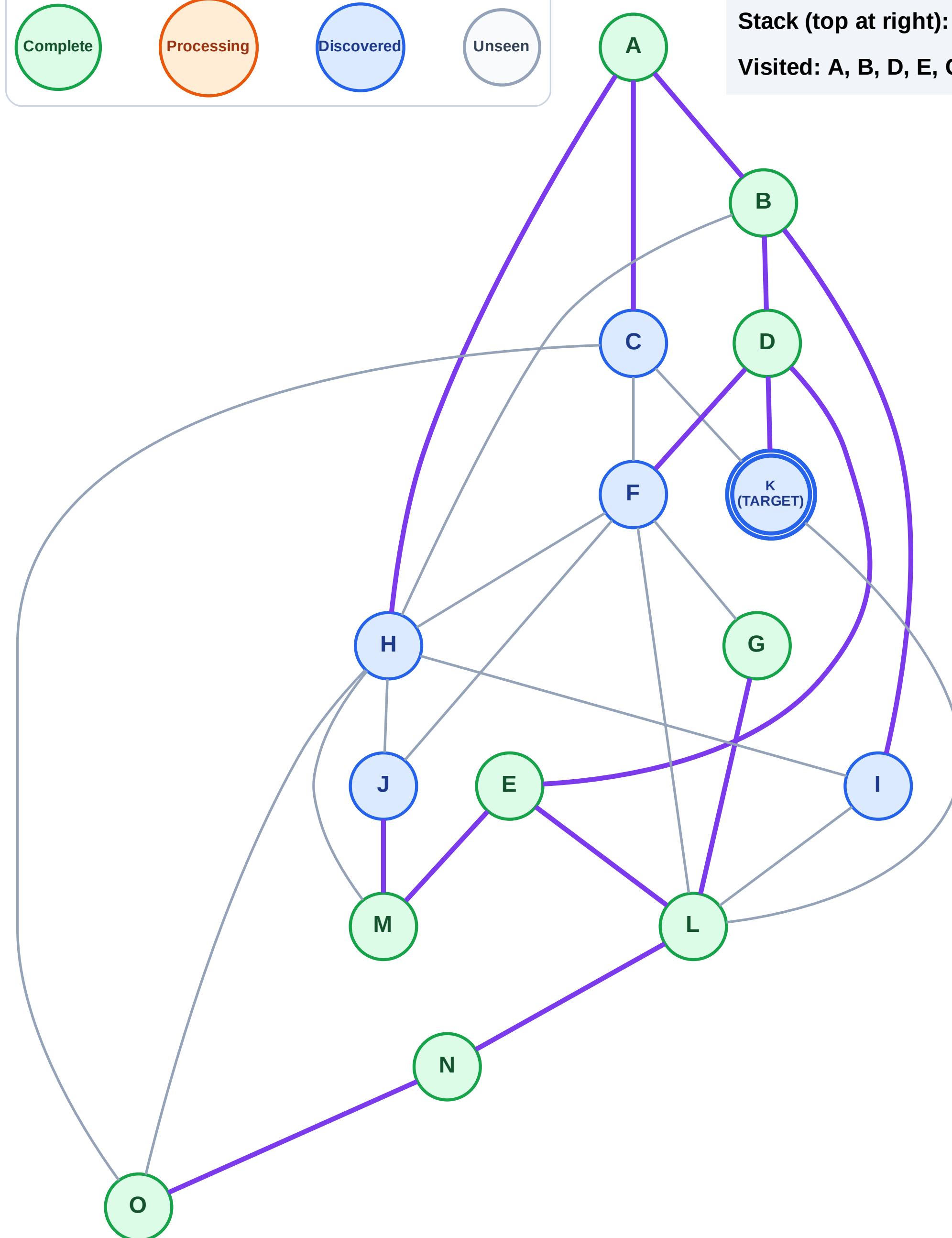
Visited: A, B, D, E, G, L, N, O



### Step 19: Discover J from M

**Stack (top at right):** H | C | I | K | F | J

**Visited:** A, B, D, E, G, L, M, N, O



# DFS Traversal

Step 20: Process node J

Stack (top at right): H | C | I | K | F  
Visited: A, B, D, E, G, L, M, N, O

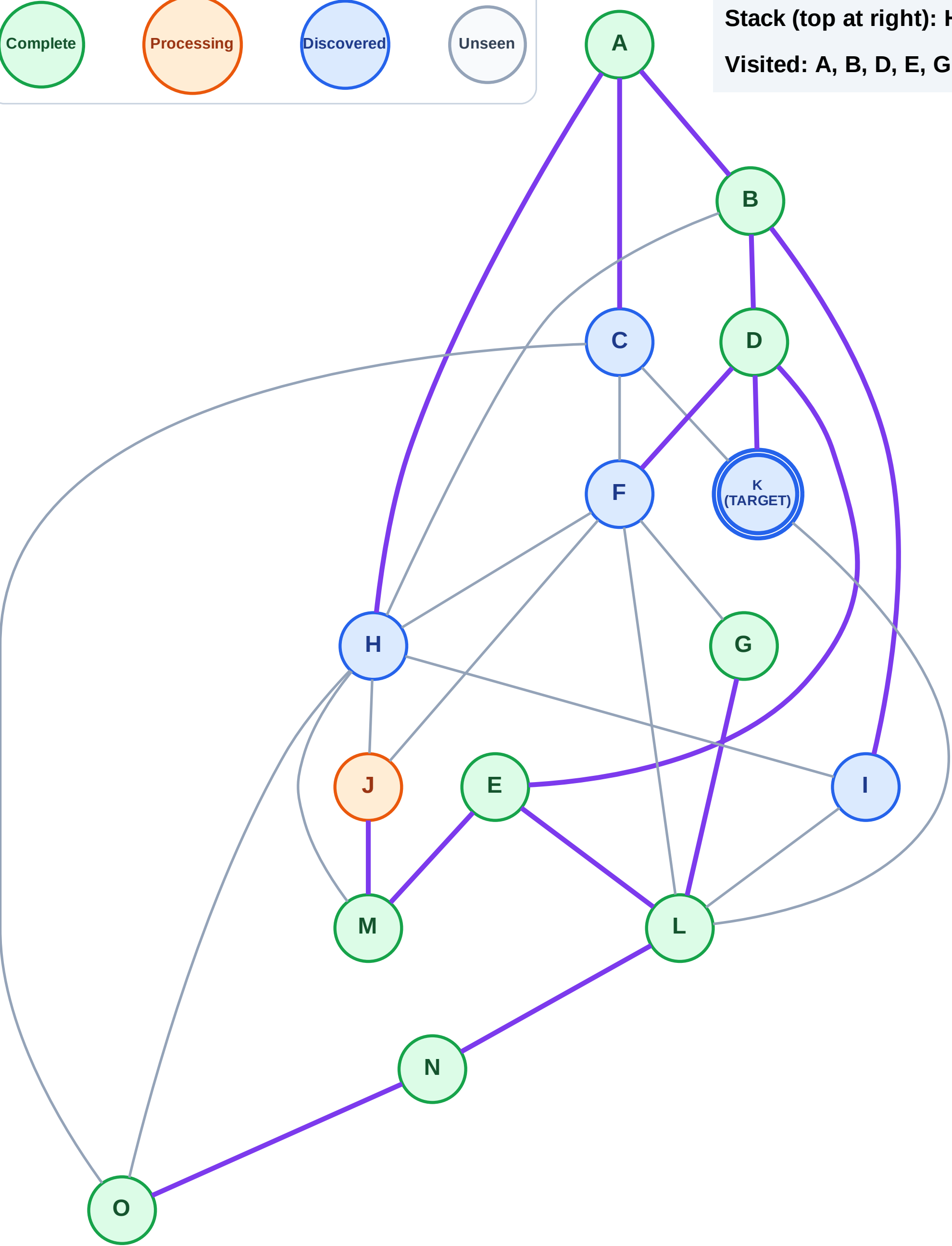
Legend

Complete

Processing

Discovered

Unseen



# DFS Traversal

Step 21: No new neighbors from J

Legend

Complete

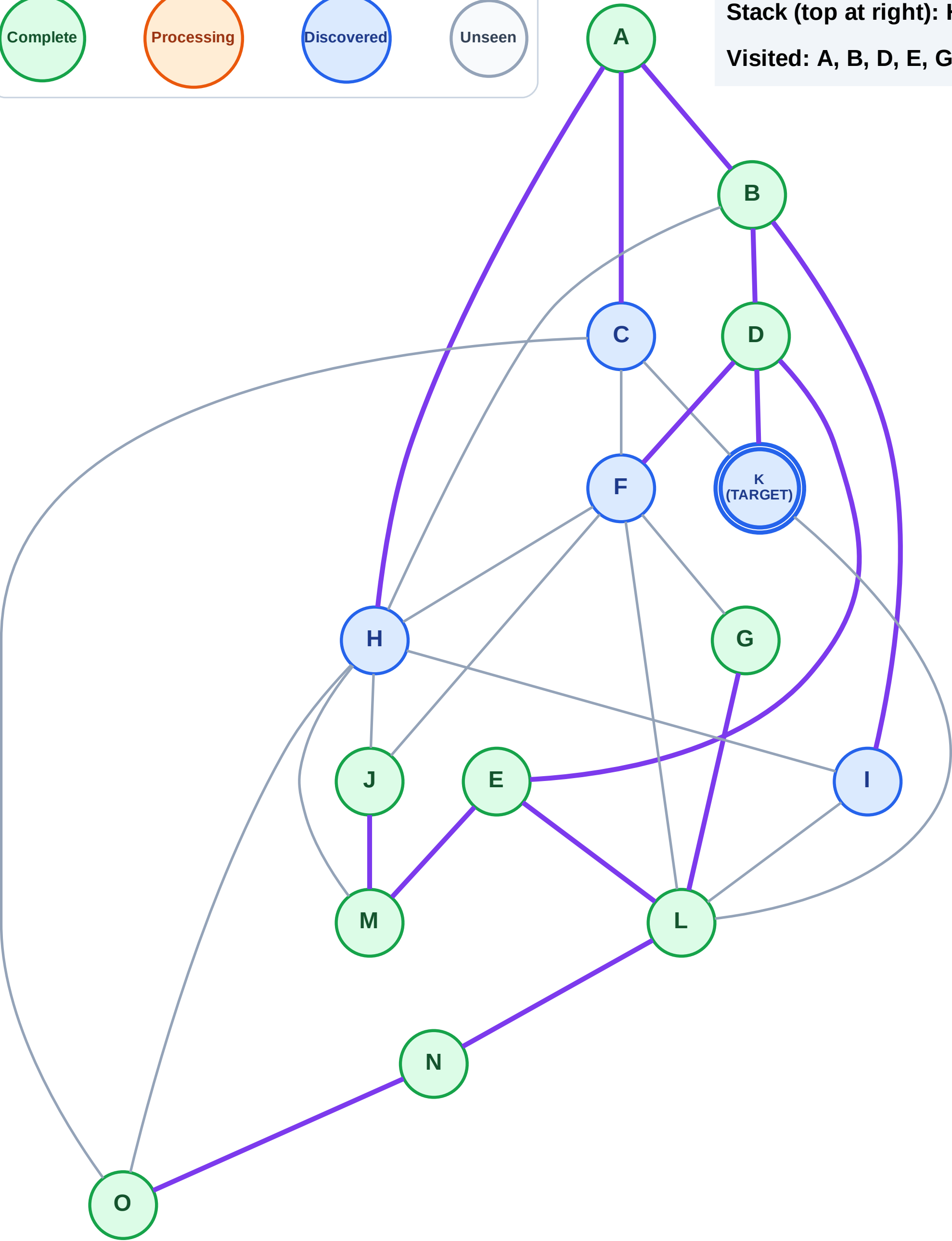
Processing

Discovered

Unseen

Stack (top at right): H | C | I | K | F

Visited: A, B, D, E, G, J, L, M, N, O



# DFS Traversal

Step 22: Process node F

Legend

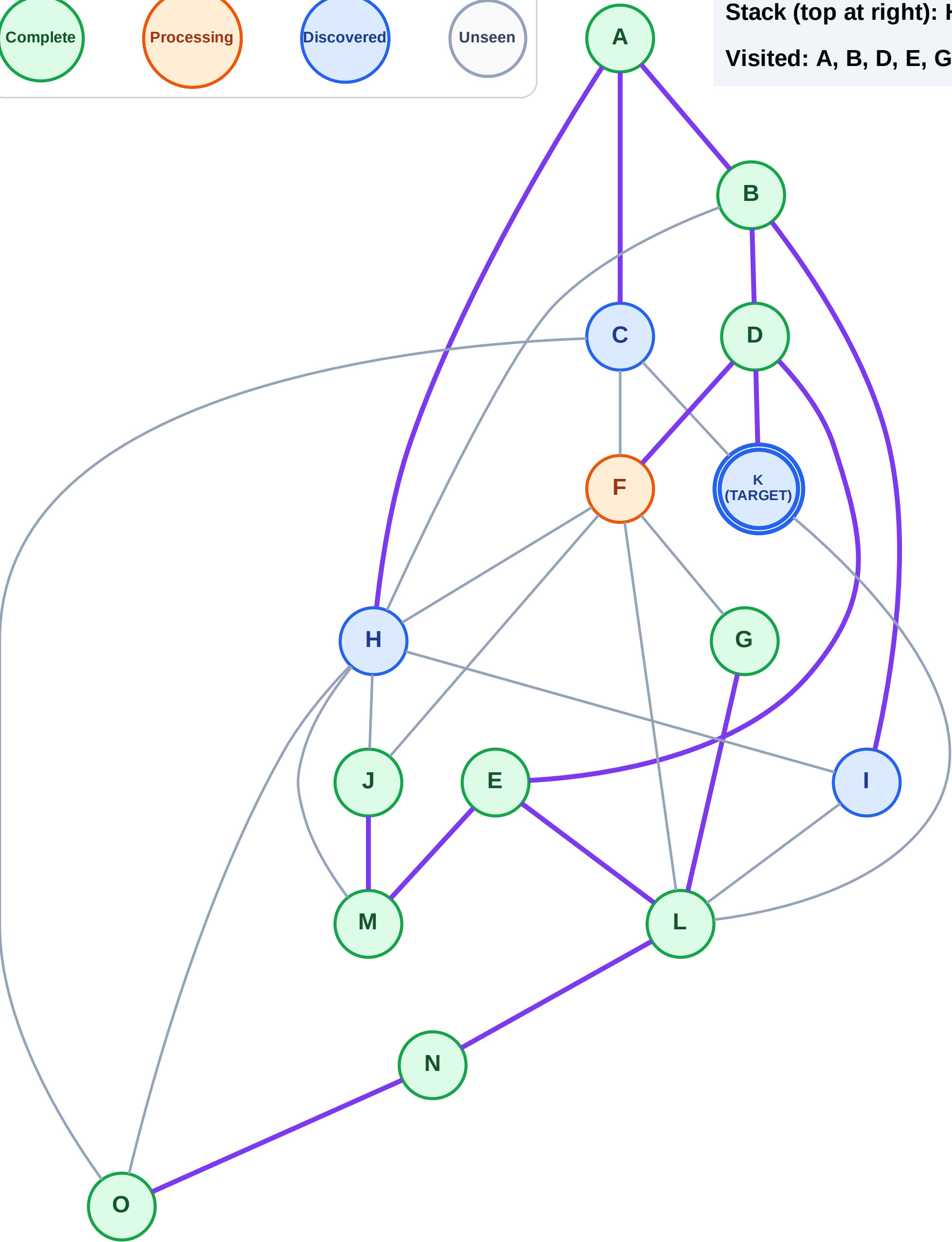
Complete

Processing

Discovered

Unseen

Stack (top at right): H | C | I | K  
Visited: A, B, D, E, G, J, L, M, N, O



# DFS Traversal

Step 23: No new neighbors from F

Legend

Complete

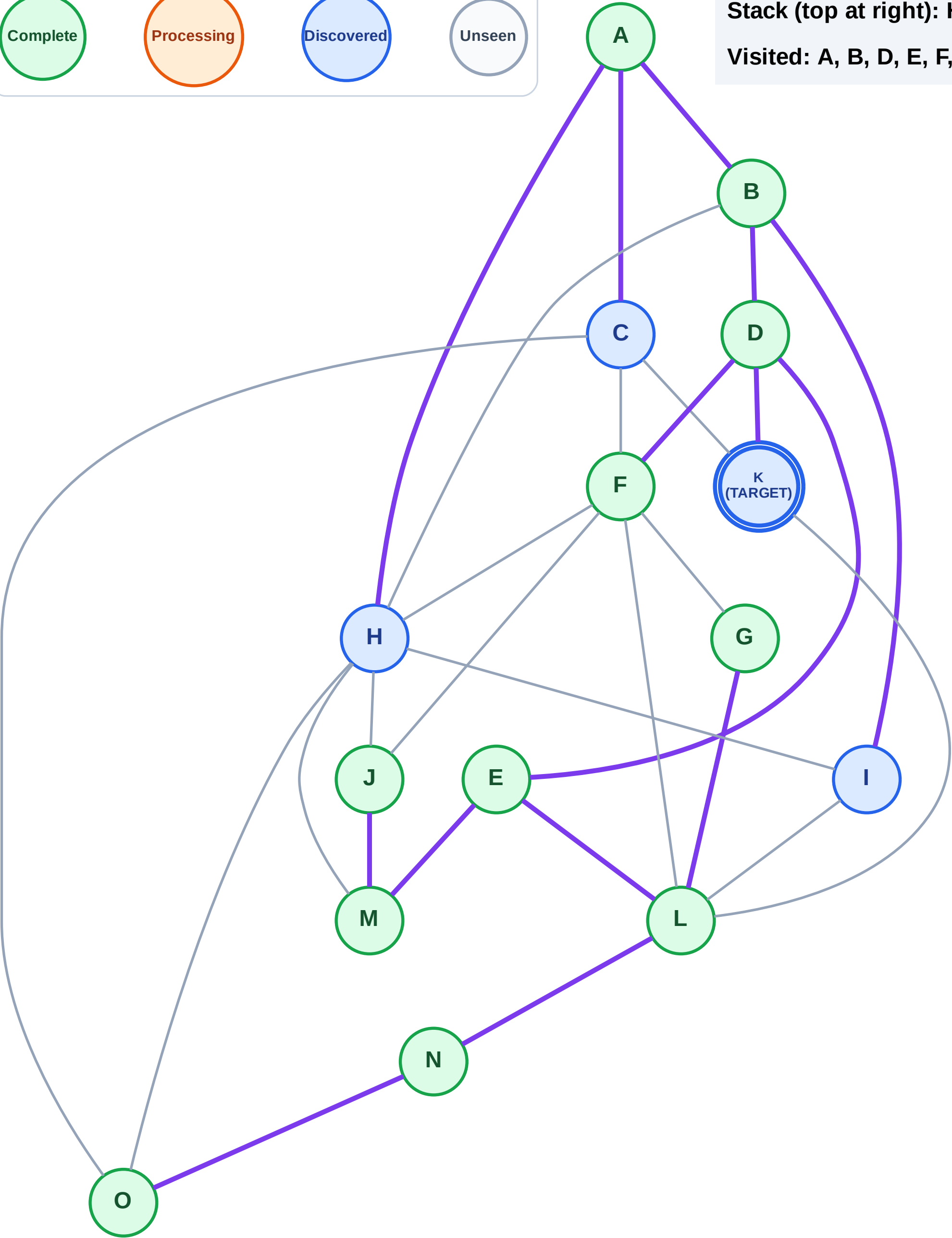
Processing

Discovered

Unseen

Stack (top at right): H | C | I | K

Visited: A, B, D, E, F, G, J, L, M, N, O



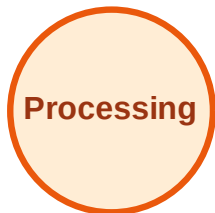
# DFS Traversal

Step 24: Process node K

## Legend



Complete



Processing



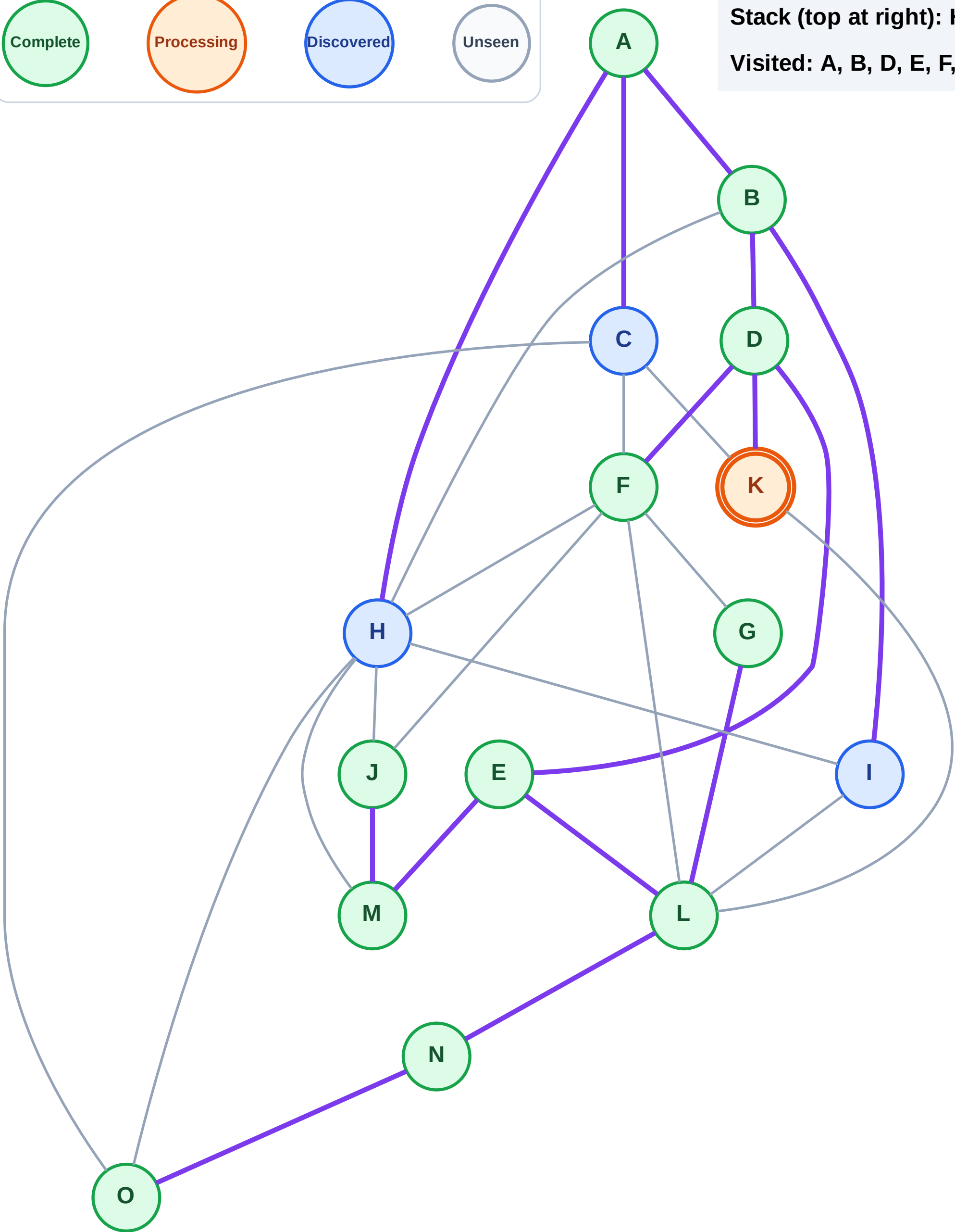
Discovered



Unseen

Stack (top at right): H | C | I

Visited: A, B, D, E, F, G, J, L, M, N, O



# DFS Traversal

Step 25: TARGET NODE K REACHED! Path (3 hops): A → B → D → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): H | C | I

Visited: A, B, D, E, F, G, J, K, L, M, N, O

